

Groundless Claims: Property Rights When Land Ceases to Exist

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Abstract

The property rights literature assumes that land continues to exist. Climate change increasingly violates this assumption. I extend the Besley (1995) framework to settings where environmental destruction permanently eliminates the asset, replacing the standard investment and credit channels with three displacement-specific mechanisms: compensation receipt, adaptation investment, and post-displacement reallocation. Calibrating the model to Kenya's Rift Valley, where rising lakes have submerged over 100 km² since 2005, I find that formally titled households recover substantially more welfare than informal occupants, with compensation accounting for the majority of the gap. Displacement amplifies pre-existing gender inequality through compounding disadvantages across all three channels, while strong customary institutions partially substitute for formal compensation. Pre-displacement titling programs yield large net benefits, particularly for women-headed households.

JEL Codes: O12, Q15, Q54, R14

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1 Introduction

Climate change is permanently destroying land. Rising seas, expanding lakes, and advancing deserts eliminate not just harvests but the asset itself, displacing millions of households whose recovery prospects depend on institutions never designed for this contingency. The Internal Displacement Monitoring Centre estimates 265 million disaster displacements between 2008 and 2018; slow-onset processes threaten far larger numbers in coming decades. Whether these households recover or fall into persistent poverty turns on how pre-existing property rights map to post-destruction outcomes – a question the economics literature has not addressed.

The reason is a gap between two mature literatures. The property rights literature, anchored by Besley (1995), shows that secure tenure raises investment, productivity, and welfare through expropriation-risk, credit, and labor-allocation channels. But the framework assumes that the land continues to exist. The disaster economics literature documents how shocks affect household welfare and recovery (Burke et al., 2015; Carleton et al., 2022; Gallagher and Hartley, 2017), but treats property rights as peripheral. To my knowledge, no peer-reviewed work connects the two.

This paper bridges the gap. I identify the implicit “land persistence” assumption in the canonical property rights framework and show that when environmental processes permanently destroy the asset, all three classic Besley channels break down. In their place, three displacement-specific channels become operative: *compensation receipt*, which depends on tenure documentation; *adaptation investment*, which depends on pre-displacement tenure security; and *post-displacement reallocation*, whose net effect is ambiguous because formalization may dissolve community-based support. I formalize these channels in a two-period household optimization model, prove seven propositions characterizing how tenure status shapes displacement welfare, and calibrate the model to generate quantitative predictions.

The calibration draws on satellite-derived lake expansion data from Kenya’s Rift Valley, where three lakes have permanently submerged over 100 km² of land since 2005, together with published estimates of tenure security gradients from sub-Saharan Africa and compensation benchmarks from disaster recovery programs. Two welfare metrics bracket the results: an accounting decomposition for channel attribution and a consumption-equivalent measure from the structural model for magnitudes. Three main findings emerge. First, formally titled households recover 82 percent of baseline welfare versus 35 percent for informal occupants (accounting) or 92 versus 77 percent (consumption-equivalent), with compensation accounting for 65 percent of the gap. Second, displacement amplifies pre-existing gender inequality by a factor of 1.8–3.1 $\times$, driven by compounding disadvantages across all three channels. Third, strong customary institutions substitute for 40–70 percent of the formal compensation gap; customary holders surpass formal holders when compensation

is equalized, revealing the insurance value of community embeddedness. These qualitative rankings hold across alternative functional forms for the claims process and in 100 percent of 2,000 Monte Carlo draws.

The paper contributes to several literatures. Most directly, it extends the property rights framework of Besley (1995) and Besley and Ghatak (2010) to settings of environmental asset destruction. Recent work has refined the standard framework – Field (2007) and Galiani and Schargrodsky (2010) show that security rather than credit drives titling effects; Goldstein and Udry (2008) demonstrate that political power determines tenure security with important gender dimensions; Adamopoulos et al. (2024) extend the literature to mobility decisions – but all maintain the land persistence assumption. Second, the paper contributes to the disaster economics literature by showing that pre-existing institutional heterogeneity, specifically tenure status, generates large and persistent differences in recovery trajectories, complementing work on compensation design (del Valle et al., 2020) and post-disaster finance (Gallagher and Hartley, 2017). Third, the gender amplification results connect the property rights and gender literatures (Goldstein and Udry, 2008; Deininger et al., 2008) to climate vulnerability, showing that displacement magnifies pre-existing inequality rather than creating new gaps.

Section 2 develops the theoretical framework. Section 3 presents the calibration strategy. Section 4 documents stylized facts from satellite data. Section 5 reports policy counterfactuals. Section 6 examines gender amplification. Appendix A provides formal proofs.

2 Theoretical Framework

This section develops an integrated framework for understanding how property rights affect household welfare in the context of environmental displacement. I begin with the canonical property rights model, identify its implicit assumption that land continues to exist, and extend the framework to incorporate environmental destruction. I then derive testable predictions for three channels through which tenure status affects displacement outcomes

and discuss heterogeneity along gender and collective tenure dimensions. The formal model and proofs are presented in Appendix A; here I develop the economic intuition.

2.1 The Standard Property Rights Framework

The foundational model of property rights and development, formalized by Besley (1995) and extended in Besley and Ghatak (2010), posits that tenure security affects economic outcomes through three primary channels:

1. **Investment channel:** Secure tenure reduces expropriation risk, increasing the expected return to land-specific investments and encouraging productivity-enhancing activities.
2. **Credit channel:** Formal title enables land to serve as collateral, relaxing credit constraints and facilitating investment.
3. **Labor allocation channel:** Security reduces the need to guard property or engage in defensive activities, freeing labor for productive uses.

Subsequent empirical work has refined this framework substantially. Field (2007) demonstrates that in Peru, titling effects operate primarily through labor reallocation rather than credit access. Galiani and Schargrotsky (2010) confirm in Argentina that security, not credit, drives investment responses. Goldstein and Udry (2008) show that in Ghana, political power determines tenure security, generating gender gaps in rights that translate into productivity differences. Most recently, Adamopoulos et al. (2024) extend the framework to show that land security affects household mobility decisions, with implications for spatial misallocation.

2.2 The Implicit Assumption: Land Persistence

A critical but unstated assumption underlies the entire property rights literature: *the land continues to exist*. The framework considers threats to tenure from expropriation, contested

inheritance, or institutional change, but not from environmental destruction of the asset itself. This assumption was reasonable when the literature developed, but climate change increasingly renders it problematic.

When land is permanently destroyed, submerged beneath rising waters, rendered uninhabitable by desertification, or swept away by erosion, the standard framework provides no guidance. The investment channel becomes irrelevant when there is nothing left to invest in. The credit channel fails when collateral ceases to exist. The labor allocation channel dissolves when there is no property left to guard. Yet the household's welfare trajectory depends critically on how pre-destruction tenure status maps to post-destruction outcomes. This is the gap the model fills.

2.3 Extending the Framework to Environmental Displacement

I propose an extension that preserves the core insight of the property rights literature, that tenure status shapes economic behavior and outcomes, while adapting it to the climate displacement context. The key innovation is recognizing that tenure status affects not only *on-land* outcomes but also *post-land* outcomes through three displacement-specific channels. The notation below corresponds to the formal model in Appendix A.

Let household i be endowed with a parcel of land of quality $q_i > 0$ and pre-displacement tenure status $T_i \in \{F, C, I\}$, representing formal title, customary rights, or informal occupation, respectively. Tenure status determines a security parameter $s_i = s(T_i) \in (0, 1]$, with $s(F) > s(C) > s(I)$.

2.3.1 Channel 1: Compensation

When an environmental shock destroys a fraction $\delta_i \in [0, 1]$ of household i 's land, formal documentation facilitates claims to government compensation, NGO assistance, and legal remedies. Let $\kappa_i(s_i, \delta_i)$ denote compensation received. Expected compensation is $\mathbb{E}[\kappa_i] = p(s_i) \cdot \bar{\kappa}(\delta_i, q_i)$, where $p(s_i)$ is the probability of a successful claim, strictly increasing in tenure security, and $\bar{\kappa}$ is the full compensation value. Formally titled households receive

larger and faster compensation because: (i) they can prove ownership and asset value; (ii) bureaucratic systems are designed around formal documentation; and (iii) they face lower transaction costs in navigating compensation processes.

Hypothesis 1 (Compensation): *Formally titled households receive compensation that is larger in magnitude and arrives more quickly than compensation received by households with customary or informal tenure.* (Proposition 1 in Appendix A.)

2.3.2 Channel 2: Adaptation

Tenure security prior to displacement affects investment in adaptation measures that reduce vulnerability. Let a_i^* denote optimal adaptation investment, and let $g(a_i, s_i)$ denote the returns to adaptation (e.g., livelihood diversification, portable assets), with $L(\delta_i, a_i)$ denoting asset losses from displacement. Following the standard investment channel logic, households with greater tenure security invest more in adaptation because they expect to capture returns over a longer horizon: the return to adaptation is higher when tenure is more secure, since the household expects to retain recognized claims, yielding the cross-partial $g_{as} > 0$. This adaptation investment reduces asset losses conditional on displacement ($\partial L / \partial a_i < 0$), so the total effect of tenure security on displacement losses is unambiguously negative.

Hypothesis 2 (Adaptation): *Households with more secure pre-displacement tenure made greater investments in adaptation measures, resulting in lower asset losses conditional on displacement.* (Propositions 2–3 in Appendix A.)

2.3.3 Channel 3: Reallocation

Tenure type affects access to alternative land and livelihood opportunities following displacement. Let N_i denote household i 's community network strength. Total reallocation value has two components: formal reallocation $R^F(s_i)$, increasing in tenure security, and community-based reallocation $R^C(N_i, T_i)$, which may be higher under customary tenure when community institutions are strong. The net effect of moving from customary to formal tenure is therefore ambiguous: formal reallocation increases but community-based

reallocation may decrease when formalization dissolves reciprocal obligations.

Hypothesis 3 (Reallocation): *Tenure type affects post-displacement land access and livelihood recovery, with formal tenure facilitating access to official programs and customary tenure enabling community-based reallocation. The net effect of formalization is ambiguous.* (Proposition 4 in Appendix A.)

2.4 Heterogeneity: Gender and Collective Tenure

Two dimensions of heterogeneity merit explicit theoretical treatment.

2.4.1 Gender

Goldstein and Udry (2008) demonstrate that women face systematically weaker tenure security due to their limited political power. In the displacement context, this generates compounding disadvantages across all three channels. Women-headed households are less likely to hold formal title, reducing compensation through Channel 1. Their weaker baseline security limits adaptation investment through Channel 2. And patrilineal inheritance norms may exclude them from community-based reallocation through Channel 3. Moreover, displacement may sever women’s tenure claims held through marriage: if a household dissolves during displacement, women may lose access to land they previously cultivated but did not formally own.

Hypothesis 4 (Gender): *Women-headed households experience worse displacement outcomes across all three channels, with effects operating through both weaker pre-displacement tenure and gender-specific barriers to compensation and reallocation. Gender gaps in post-displacement welfare exceed gender gaps in pre-displacement welfare: $\Delta W^{post} > \Delta W^{pre}$.* (Proposition 5 in Appendix A.)

2.4.2 Collective Tenure

The literature following Ostrom (1990) and extended by Montero (2022) demonstrates that collective property arrangements can function efficiently when governance institutions are

well-designed. In the displacement context, collective tenure creates both vulnerabilities and resilience mechanisms. Collective tenure may complicate compensation claims, as individual households cannot document individual losses. However, collective institutions may facilitate community-level reallocation and mutual insurance transfers. Formally, customary tenure provides partial insurance against displacement losses through community transfer mechanisms, with the degree of insurance increasing in institutional quality θ . There exists a threshold $\bar{\theta}$ above which community insurance fully compensates for the formal compensation gap.

Hypothesis 5 (Collective Tenure): *Households embedded in strong collective tenure institutions experience partial substitution of community-based support for formal compensation, with net effects depending on institutional quality.* (Proposition 6 in Appendix A.)

2.5 Welfare Decomposition

Combining the three channels, post-displacement welfare admits the decomposition:

$$W_i^{post} = W_i^{pre} - L(\delta_i, a_i^*(s_i)) + \kappa_i(s_i) + R_i(T_i, N_i) \quad (1)$$

where losses L are decreasing in tenure security (through adaptation), compensation κ is increasing in tenure security, and reallocation R is ambiguous. The total effect of tenure security on post-displacement welfare satisfies $dW_i^{post}/ds_i > 0$ when the compensation and loss-reduction channels dominate the potentially negative reallocation effect – a condition that holds when formal compensation mechanisms exist (Proposition 7 in Appendix A).

The framework generates six testable predictions: (1) compensation receipt is increasing in tenure formality; (2) pre-displacement adaptation investment is increasing in tenure security; (3) asset losses are decreasing in prior adaptation; (4) women-headed households experience worse outcomes across all dimensions; (5) customary institutions provide partial insurance; and (6) the gap in outcomes between tenure types is larger when formal compensation mechanisms are weak or delayed. The calibration exercise in Section 3 quantifies

the magnitude of each channel.

3 Calibration Strategy

3.1 Model Summary

The theoretical framework in Section 2 develops the economic intuition; here I state the formal optimization problem for calibration. Full proofs of all propositions are provided in Appendix A.

Setup

Consider a household i endowed with a parcel of land of quality $q_i > 0$ and tenure status $T_i \in \{F, C, I\}$ (formal, customary, informal). Tenure status determines a security parameter $s_i = s(T_i) \in (0, 1]$, where $s(F) > s(C) > s(I)$. The parameter s_i captures the probability that the household retains recognized claims to its land and land-linked entitlements. Time is discrete with two periods: $t = 0$ (pre-displacement) and $t = 1$ (post-displacement). At $t = 0$, the household chooses investment in productive capital k_i and adaptation capital a_i subject to a budget constraint. At $t = 1$, an environmental shock destroys a fraction $\delta_i \in [0, 1]$ of the land, and the household receives compensation and accesses reallocation opportunities.

Household Optimization Problem

At $t = 0$, the household solves:

$$\max_{k_i, a_i} U_i = u(c_i^0) + \beta \mathbb{E} [u(c_i^1)] \quad (2)$$

subject to the period-0 budget constraint $c_i^0 = y_i - k_i - a_i - \phi(s_i)$, where y_i is the household's period-0 endowment (normalized to 1 in the calibration) and $\phi(s_i) = \phi_0(1 - s_i)$ is a documentation cost decreasing in tenure security. Period-1 consumption depends on

displacement severity:

$$c_i^1 = f((1 - \delta_i \cdot \ell(a_i, s_i)) q_i, k_i) + g(a_i, s_i) + p(s_i)\bar{\kappa} \quad (3)$$

where $f(\cdot)$ is a Cobb-Douglas production function operating on effective land (reduced by displacement severity δ_i and moderated by the asset protection function $\ell(a_i, s_i) = e^{-\lambda a_i s_i}$), $g(a_i, s_i) = \gamma a_i^\eta s_i$ captures returns to adaptation investment, and $p(s_i)\bar{\kappa}$ is expected compensation. The reallocation and community transfer components (R_i and Ω_i) enter the accounting welfare decomposition but are not part of the household's optimization problem, as they are realized ex post.

Propositions

The model yields seven propositions, stated below. Proofs are in Appendix A.

- P1. Compensation Channel.** Expected compensation $\mathbb{E}[\kappa_i]$ is strictly increasing in tenure security s_i , and the expected time to compensation receipt is strictly decreasing in s_i .
- P2. Adaptation Investment.** Optimal adaptation investment a_i^* is strictly increasing in tenure security s_i , since higher security raises the expected return to adaptation through the cross-partial $g_{as} > 0$.
- P3. Adaptation Reduces Losses.** Asset losses from displacement are strictly decreasing in adaptation investment ($\partial L / \partial a_i < 0$). Combined with P2, total losses are decreasing in tenure security.
- P4. Reallocation Channel.** The effect of moving from customary to formal tenure on total reallocation value is ambiguous: formal reallocation increases ($\partial R^F / \partial s > 0$) but community-based reallocation may decrease when formalization dissolves reciprocal obligations.

P5. Gender Amplification. If women face systematically lower tenure security $s(T|W) < s(T|M)$ for all T , and displacement can sever tenure claims held through marriage, then gender gaps in post-displacement welfare exceed gender gaps in pre-displacement welfare: $\Delta W^{post} > \Delta W^{pre}$.

P6. Collective Insurance. Customary tenure provides partial insurance against displacement losses through community transfer mechanisms, with the degree of insurance increasing in institutional quality θ . There exists a threshold $\bar{\theta}$ above which community insurance fully compensates for the formal compensation gap.

P7. Welfare Decomposition. Post-displacement welfare admits the decomposition $W_i^{post} = W_i^{pre} - L(\delta_i, a_i^*(s_i)) + \kappa_i(s_i) + R_i(T_i, N_i)$, with $dW_i^{post}/ds_i > 0$ when compensation and loss-reduction channels dominate the potentially negative reallocation effect.

3.2 Model Parameters

The remainder of this section describes how to parameterize the model developed in Sections 2–3.1 (proofs in Appendix A) to generate quantitative predictions. The calibration draws on three sources: (i) satellite-derived environmental data from the GEE pipeline, (ii) published microeconomic estimates from sub-Saharan Africa, and (iii) compensation benchmarks from disaster recovery programs.

The model requires calibrating the following parameter groups:

3.3 Environmental Destruction Parameter: δ_i

The key advantage of this setting is that δ_i is *directly observable from satellite data*. Unlike models of political risk to property (He et al., 2024), where the threat of expropriation is latent, lake expansion physically destroys land. I construct household-level δ_i as follows.

Table 1: Model Parameters and Calibration Sources

Parameter	Description	Value/Range	Source
<i>Panel A: Environmental Destruction</i>			
δ_i	Displacement severity (land fraction lost)	0–0.84	Satellite data (Baringo)
$\dot{\delta}$	Annual expansion rate	0.02–0.12	Lake area time series
π_i	Probability of displacement	0.10–0.95	Distance-to-lake bins
<i>Panel B: Tenure Security</i>			
$s(F)$	Security under formal title	0.85–0.95	Subramanian and Kumar (2024)
$s(C)$	Security under customary tenure	0.50–0.70	Goldstein and Udry (2008)
$s(I)$	Security under informal occupation	0.10–0.30	Adamopoulos et al. (2024)
<i>Panel C: Compensation Process</i>			
$p(s)$	Claim success probability	$p(s) = s^\alpha$	Functional form assumption
α	Concavity of claims process	0.5–1.0	Calibrated
$\bar{\kappa}$	Full compensation value (% of land)	0.40–0.80	del Valle et al. (2020)
$\phi(s)$	Documentation cost	$\phi_0(1 - s)$	Manara and Regan (2025)
<i>Panel D: Adaptation Technology</i>			
$g(a, s)$	Returns to adaptation	$\gamma a^\eta s$	CES specification
γ	Adaptation productivity	0.3–0.8	Carleton et al. (2022)
η	Diminishing returns	0.4–0.6	Standard concavity
$\ell(a)$	Asset protection function	$e^{-\lambda a}$	Exponential decay
λ	Protection effectiveness	0.5–2.0	Calibrated to match losses
<i>Panel E: Household Fundamentals</i>			
β	Discount factor	0.90–0.95	Standard (annual)
$u(c)$	Utility	$c^{1-\sigma}/(1-\sigma)$	CRRA
σ	Risk aversion	1.5–3.0	Lagakos et al. (2023)
q_i	Land quality (hectares)	0.5–5.0	KIHBS median
$f(q, k)$	Production function	$Aq^{\alpha_q}k^{\alpha_k}$	Cobb-Douglas
<i>Panel F: Community Insurance</i>			
θ	Institutional quality	0–1	Index (to be constructed)
V^{cont}	Continuation value of insurance	Endogenous	Solved in equilibrium
n_d/n	Fraction of community displaced	0.05–0.50	Satellite $\times$ settlement data
μ	Marital claim severance probability	0.15–0.30	Qualitative evidence

3.3.1 From Lake Area to Household Displacement Severity

Define A_t^{lake} as the satellite-measured lake surface area in year t . The expansion zone at time t is:

$$E_t = A_t^{lake} - A_0^{lake} \quad (4)$$

where A_0^{lake} is the baseline (2005) area. From the satellite data:

Table 2: Observed Lake Expansion (Satellite Data)

Lake	Baseline 2005 (km ²)	Peak Area (km ²)	Peak Year	Expansion (km ²)	Displacement Intensity
Baringo	121.0	222.9	2025	101.9	0.843
Bogoria	11.5	34.5	2025	23.1	2.012
Naivasha	122.1	162.7	2025	40.7	0.333

For a household i located at distance d_i from the 2005 lake boundary, displacement severity is:

$$\delta_{it} = \begin{cases} 1 & \text{if } d_i \leq r_t \text{ (fully inundated)} \\ \max\{0, 1 - (d_i - r_t)/\bar{d}\} & \text{if } d_i \in (r_t, r_t + \bar{d}) \text{ (partial disruption)} \\ 0 & \text{if } d_i > r_t + \bar{d} \text{ (unaffected)} \end{cases} \quad (5)$$

where r_t is the expansion radius at time t (derived from E_t under approximate radial symmetry) and $\bar{d}$ is a disruption buffer (calibrated at 2–5 km for infrastructure, groundwater, and access effects).

3.3.2 Temporal Profile of δ

A key feature distinguishing slow-onset from sudden-onset displacement is the *gradual accumulation* of δ . From the satellite time series, I characterize three phases:

1. **Creeping phase (2005–2010):** $\dot{\delta} \approx 0.02$ per year. Baringo expanded ~ 12 km² above baseline. Households observe rising water but displacement is minimal.

2. **Acceleration phase (2011–2015):** $\dot{\delta} \approx 0.08$ per year. Baringo expanded $\sim 65 \text{ km}^2$ above baseline. Mass displacement begins. This is when adaptation decisions are most consequential – households that invested early in movable assets or elevated structures preserve more wealth.
3. **Plateau phase (2017–2025):** $\dot{\delta} \approx 0.03$ per year. Expansion continues but decelerates. Cumulative displacement intensity reaches 0.84 at Baringo. Compensation and reallocation mechanisms are active.

This temporal profile is critical for the model because it determines the *information environment* for adaptation decisions. Households observing the creeping phase face genuine uncertainty about whether the trend will continue, making the option value of adaptation investment sensitive to tenure security (through the $g_{as} > 0$ cross-partial in Proposition 2).

3.4 Tenure Security Parameters: $s(F)$, $s(C)$, $s(I)$

The model requires cardinal values for tenure security across three regimes. I draw on recent empirical estimates:

1. **Formal title, $s(F) = 0.90$:** Subramanian and Kumar (2024) find that titled households in their India RCT report 90 percent perceived probability of retaining land over a 10-year horizon. Manara and Regan (2025) find similar magnitudes in Tanzania where formal titling raised perceived security from ~ 0.65 to ~ 0.90 .
2. **Customary tenure, $s(C) = 0.60$:** Goldstein and Udry (2008) document that tenure security in Ghanaian customary systems ranges from 0.45 (women without political connections) to 0.75 (men with connections). I use 0.60 as a midpoint. This is consistent with Kenyan evidence on community land tenure.
3. **Informal occupation, $s(I) = 0.20$:** Adamopoulos et al. (2024) show that households without recognized claims face very high expropriation risk. Galiani and Schargrodsky (2010) estimate baseline security of ~ 0.15 – 0.25 for Buenos Aires squatters pre-titling.

Sensitivity: I vary these across ± 0.10 to check robustness. The key qualitative results depend on the *ordering* $s(F) > s(C) > s(I)$ rather than exact magnitudes.

3.5 Compensation Process Parameters

3.5.1 Functional Form: $p(s) = s^\alpha$

The claim success probability $p(s)$ must satisfy $p'(s) > 0$, $p''(s) \leq 0$, $p(0) = 0$, and $p(1) = 1$. The power function $p(s) = s^\alpha$ with $\alpha \in (0, 1]$ satisfies all conditions. Under this specification:

$$p(s(F)) = 0.90^{0.7} \approx 0.93 \tag{6}$$

$$p(s(C)) = 0.60^{0.7} \approx 0.70 \tag{7}$$

$$p(s(I)) = 0.20^{0.7} \approx 0.32 \tag{8}$$

This implies formally titled households have a 93 percent chance of successful compensation claims, customary tenure holders 70 percent, and informal occupants 32 percent. These magnitudes are consistent with anecdotal evidence from Kenya’s displacement compensation programs, where documented owners received priority.

3.5.2 Compensation Value: $\bar{\kappa}$

Full compensation as a fraction of land value: del Valle et al. (2020) study FEMA buyout programs and find compensation averaging 60–80 percent of pre-disaster market value. Kenya’s National Land Commission benchmarks are lower; I use $\bar{\kappa} = 0.50$ (50 percent of asset value) as baseline with range 0.30–0.80.

3.5.3 Expected Compensation by Tenure Type

Under the calibration:

$$\mathbb{E}[\kappa|F] = p(s(F)) \cdot \bar{\kappa} = 0.93 \times 0.50 = 0.465 \quad (9)$$

$$\mathbb{E}[\kappa|C] = p(s(C)) \cdot \bar{\kappa} = 0.70 \times 0.50 = 0.350 \quad (10)$$

$$\mathbb{E}[\kappa|I] = p(s(I)) \cdot \bar{\kappa} = 0.32 \times 0.50 = 0.162 \quad (11)$$

Prediction 1: Formally titled households receive $\sim 3\times$ the compensation of informal occupants, and $\sim 1.4\times$ that of customary holders.

3.5.4 Expected Wait Time

From the geometric claims process (Proposition 1, Part 2):

$$\mathbb{E}[\tau|F] = 1/0.93 \approx 1.1 \text{ rounds} \quad (12)$$

$$\mathbb{E}[\tau|C] = 1/0.70 \approx 1.4 \text{ rounds} \quad (13)$$

$$\mathbb{E}[\tau|I] = 1/0.32 \approx 3.1 \text{ rounds} \quad (14)$$

If each “round” corresponds to ~ 6 months of bureaucratic processing, formal title holders wait ~ 7 months, customary holders ~ 9 months, and informal occupants ~ 19 months.

Prediction 2: Informal occupants wait $\sim 2.8\times$ longer than titled households for compensation.

3.5.5 Robustness to Alternative Functional Forms

Because compensation accounts for 65 percent of the formal-informal welfare gap (Section 3), the results are sensitive to the specification of $p(s)$. I check robustness by re-solving the model under two alternative forms:

- **Logistic:** $p(s) = 1/(1 + e^{-10(s-0.5)})$, which concentrates the claims gradient around

a threshold at $s = 0.5$. This captures a documentation regime where claims succeed abruptly once tenure crosses a formality threshold.

- **Linear with floor:** $p(s) = \max(0.05, s)$, which is proportional to security with a small floor ensuring that even fully informal households retain some probability of compensation.

Table 3 reports the welfare decomposition under each specification. All three forms preserve the qualitative ordering $W^{post}(F) > W^{post}(C) > W^{post}(I)$. The logistic form produces the largest formal-informal gap (CE gap of 0.228) because the threshold at $s = 0.5$ sharply penalizes informal occupants ($p(0.20) = 0.047$, vs. 0.324 under the power form). The linear form yields an intermediate gap (CE = 0.179). The baseline power function is the most conservative specification for the formal-informal gap.

Table 3: Robustness to Claims Process Functional Form ($\delta = 0.84$)

Form	Tenure	$p(s)$	Comp.	W^{post}	CE λ
Power $s^{0.7}$ (baseline)	F	0.929	0.464	0.816	0.917
	C	0.699	0.350	0.733	0.874
	I	0.324	0.162	0.348	0.768
Logistic $1/(1 + e^{-10(s-0.5)})$	F	0.982	0.491	0.837	0.920
	C	0.731	0.366	0.746	0.877
	I	0.047	0.024	0.215	0.693
Linear $\max(0.05, s)$	F	0.900	0.450	0.804	0.915
	C	0.600	0.300	0.691	0.863
	I	0.200	0.100	0.288	0.737
F-I gap (CE)			Power: 0.149	Logistic: 0.228	Linear: 0.179

The key takeaway is that the choice of functional form affects the *magnitude* of the formal-informal gap but not its *existence or sign*. The power specification, which produces the smallest gap, is therefore a conservative baseline. If the true claims process involves threshold effects – as documentation requirements in bureaucratic systems plausibly do – the welfare costs of informal tenure are larger than the baseline calibration suggests.

3.6 Adaptation Technology Parameters

3.6.1 Returns to Adaptation: $g(a, s) = \gamma a^\eta s$

Adaptation investment a serves two functions in the model: it reduces asset losses through the protection function $\ell(a, s)$ (on-land adaptation such as raised structures and drainage), and it generates direct consumption returns $g(a, s)$ that persist post-displacement (portable adaptation such as livelihood diversification, human capital investment, and acquisition of movable assets). The separability of g from land is intentional: in the Rift Valley context, households that invested pre-displacement in livestock relocation, skills training, or non-agricultural enterprises retained those returns even after their land was submerged. The interaction with tenure security ($g_{as} > 0$) reflects the fact that such forward-looking investments require a secure planning horizon – households uncertain about their tenure are less likely to invest in adaptation of any kind.

The return to adaptation investment depends on both the quality of adaptation and tenure security (Proposition 2 requires $g_{as} > 0$). With $\gamma = 0.5$ and $\eta = 0.5$:

For a household investing a out of a normalized unit endowment:

$$g(a, s(F)) = 0.5 \cdot a^{0.5} \cdot 0.90 = 0.45 a^{0.5} \quad (15)$$

$$g(a, s(C)) = 0.5 \cdot a^{0.5} \cdot 0.60 = 0.30 a^{0.5} \quad (16)$$

$$g(a, s(I)) = 0.5 \cdot a^{0.5} \cdot 0.20 = 0.10 a^{0.5} \quad (17)$$

3.6.2 Optimal Adaptation Investment

Under log utility the closed-form approximation is $a^*(s) \propto (\beta\gamma\eta s / (1 + \beta\gamma\eta s))^{1/(1-\eta)}$, but with CRRA utility ($\sigma = 2$) and the full two-period problem the first-order condition is non-linear. I solve numerically using `scipy.optimize` (see `01_code/calibration/calibrate_model.py`).

Under the baseline calibration ($\beta = 0.92$, $\gamma = 0.5$, $\eta = 0.5$, $\sigma = 2$, $\lambda = 1.5$):

$$a^*(F) = 0.095 \quad (9.5\% \text{ of endowment}) \quad (18)$$

$$a^*(C) = 0.087 \quad (8.7\% \text{ of endowment}) \quad (19)$$

$$a^*(I) = 0.025 \quad (2.5\% \text{ of endowment}) \quad (20)$$

Figure 1 plots $a^*(s)$ over the continuous support, with vertical markers at the three tenure benchmarks.

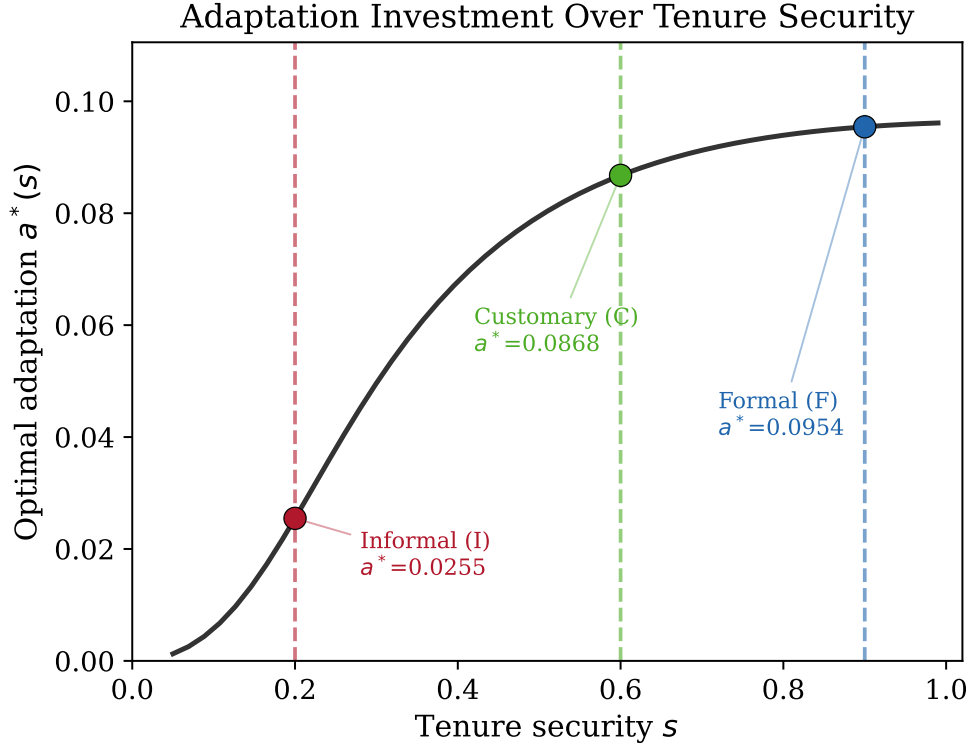


Figure 1: Optimal adaptation investment as a function of tenure security, solved numerically under CRRA ($\sigma = 2$). Vertical dashed lines mark the calibrated security values for formal (F), customary (C), and informal (I) tenure.

Prediction 3: Formally titled households invest $\sim 1.1\times$ more in adaptation than customary holders and $\sim 3.8\times$ more than informal occupants. The formal-customary gap is small, consistent with similar incentives for asset protection; the customary-informal gap is large, reflecting the sharp drop in adaptation returns at low s .

3.6.3 Asset Protection Function: $\ell(a, s) = e^{-\lambda as}$

The fraction of at-risk assets destroyed decreases exponentially in both adaptation and tenure security: $\ell(a, s) = e^{-\lambda as}$, where the interaction captures the idea that protective investments require legal backing to be effective. With $\lambda = 1.5$ and numerically solved a^* :

$$\ell(a^*(F), s(F)) = e^{-1.5 \times 0.095 \times 0.90} = 0.879 \quad (21)$$

$$\ell(a^*(C), s(C)) = e^{-1.5 \times 0.087 \times 0.60} = 0.925 \quad (22)$$

$$\ell(a^*(I), s(I)) = e^{-1.5 \times 0.025 \times 0.20} = 0.993 \quad (23)$$

Multiplying by displacement severity $\delta = 0.84$, the asset loss fractions are:

$$L(F) = 0.84 \times 0.879 = 0.739 \quad (24)$$

$$L(C) = 0.84 \times 0.925 = 0.777 \quad (25)$$

$$L(I) = 0.84 \times 0.993 = 0.834 \quad (26)$$

The 9.5 percentage point gap between formal and informal losses ($L(F) = 0.739$ vs. $L(I) = 0.834$) reflects both (i) higher adaptation investment and (ii) more effective protection per unit invested. This gap is amplified through the compensation and reallocation channels below.

3.7 Production Function Calibration

The Cobb-Douglas specification $f(q, k) = Aq^{\alpha_q}k^{\alpha_k}$ is calibrated using Kenyan agricultural data. I set the land share $\alpha_q = 0.4$ and the capital share $\alpha_k = 0.5$, consistent with estimates from smallholder agriculture in sub-Saharan Africa (Restuccia and Santaaulàlia-Llopis, 2017). TFP is normalised to $A = 1$ and land quality to $q = 1$, so that production is scaled relative to the household endowment.

Under displacement with severity $\delta = 0.84$, effective land falls to $q(1 - \delta \cdot \ell(a^*, s))$. For formally titled households, adaptation reduces the effective land loss to $q \times 0.261$ (vs.

$q \times 0.166$ for informal occupants), meaning formal households retain 26 percent of productive capacity compared to 17 percent for informal occupants. Combined with higher capital investment ($k^*(F) = 0.087$ vs. $k^*(I) = 0.273$), this generates the consumption gap documented in Table 4.

The high capital investment by informal households ($k^*(I) = 0.273$) reflects a portfolio reallocation from adaptation to capital: absent effective adaptation returns (because s is low), these households shift resources toward physical capital as a substitute, but the returns are low because effective land is severely reduced.

3.8 Reallocation Calibration

The reallocation component R_i is parameterized as $R(T) = 0.10 \times s(T)$, yielding $R(F) = 0.090$, $R(C) = 0.060$, and $R(I) = 0.020$. The linear specification reflects the assumption that access to post-displacement land and livelihood opportunities scales with tenure security – formally titled households can access government resettlement programs, use documentation to purchase or lease new land, and leverage legal standing in negotiations, while informal occupants face barriers at each step.

The coefficient of 0.10 is calibrated to yield a modest contribution relative to compensation and asset losses, consistent with evidence from IDP resettlement programs in sub-Saharan Africa where formal land access post-displacement is limited. The UNDP (2021) rapid assessment of Kenya’s Rift Valley displacement found that fewer than 15 percent of displaced households had accessed formal resettlement, with most relying on kinship networks and informal arrangements – motivating reallocation’s small share (15 percent) of the total formal-informal gap. To verify that the results do not depend on this parameter, I note that setting $R = 0$ for all tenure types reduces the formal-informal accounting gap from 0.468 to 0.398 (a 15 percent reduction) while the CE gap falls from 0.149 to 0.127, preserving the qualitative ordering and all key predictions.

Community transfers for customary holders ($\Omega = \theta \cdot \omega_{max}$) operate as a separate insurance mechanism through Proposition 6, with $\omega_{max} = 0.20$ (the maximum community

transfer as a fraction of welfare, sustained by participation constraints). The 20 percent ceiling is anchored in the village-level risk-sharing literature: Townsend (1994) finds that villages in southern India achieve 70–80 percent consumption insurance, implying net transfers of approximately 20–30 percent of income among affected households. Montero (2022) documents that collective tenure arrangements in El Salvador sustained comparable transfer rates during land reform disruptions. In the Kenyan context, community transfers are bounded from above by the fraction of the community that remains undisplaced ($(n - n_d)/n_d$ in Proposition 6), which limits the sustainable per-recipient transfer when displacement is concentrated. The sensitivity table (Table 6) shows the full range of outcomes under $\theta \in [0, 1]$: at $\theta = 0$ (no community insurance), customary welfare falls to 0.633; at $\theta = 1$, it rises to 0.833.

3.9 Welfare Decomposition: Putting It Together

I present two complementary welfare metrics. The first is an *accounting decomposition* that traces post-displacement welfare through each model channel, providing transparency about where the formal-informal gap originates. The second is a *consumption-equivalent* measure derived from the utility-based solution of the two-period model, which disciplines the magnitudes by respecting the concavity of preferences.

Accounting Decomposition

Normalize baseline welfare $W^{pre} = 1$ for all tenure types. Under displacement severity $\delta = 0.84$ at Baringo, the channel-by-channel decomposition is:

The accounting decomposition is an additive framework that uses the model to determine only two endogenous objects: optimal adaptation $a^*(s)$ from the household solver and the resulting asset protection $\ell(a^*, s)$. The remaining components, compensation $p(s)\bar{\kappa}$, community transfers Ω , and reallocation R , are evaluated directly from the calibrated parameters. This linearity is the decomposition’s strength: it makes the channel contributions transparent and comparable. Its limitation is that it treats each channel as contributing

Table 4: Welfare Decomposition: Accounting Framework ($\delta = 0.84, \sigma = 2$)

Component	Formal (F)	Customary (C)	Informal (I)
Baseline welfare W^{pre}	1.000	1.000	1.000
Asset losses $-L(\delta, a^*, s)$	-0.739	-0.777	-0.834
Compensation $\kappa(s)$	+0.465	+0.350	+0.162
Community transfer Ω	+0.000	+0.100	+0.000
Reallocation $R(T)$	+0.090	+0.060	+0.020
Accounting recovery W^{post}	0.816	0.733	0.348

independently to welfare, abstracting from the concavity of utility. For parameter combinations where compensation and reallocation are large relative to losses, the accounting decomposition can exceed 1 (i.e., imply post-displacement welfare above baseline), which is economically implausible. I therefore cap the accounting measure at $W^{post} \leq W^{pre}$ and use the consumption-equivalent metric below as a structural cross-check.

Consumption-Equivalent Welfare

The structural welfare metric is the consumption-equivalent ratio λ : the fraction of no-displacement consumption a household would accept in exchange for avoiding the displacement lottery. Formally, λ solves $U(\lambda \cdot c^{pre}) = U(c^{post})$, where U is the solved two-period CRRA objective. Under the baseline calibration:

Table 5: Consumption-Equivalent Welfare ($\delta = 0.84, \sigma = 2$)

	Formal (F)	Customary (C)	Informal (I)
CE welfare λ	0.917	0.874	0.768
CE recovery gap ($F - I$)		0.149	
CE F/I ratio		1.19×	

The CE metric confirms the qualitative ranking $\lambda(F) > \lambda(C) > \lambda(I)$ but yields a more modest F/I ratio of 1.19× compared to the accounting framework’s 2.34×. The difference reflects the concavity of CRRA utility: large accounting losses translate into smaller utility losses because households are already on the steep portion of the utility function, and the gains from compensation are similarly compressed. The CE measure is bounded above by 1 by construction, avoiding the over-unity problem of the accounting framework.

Figure 2 presents the accounting decomposition visually.

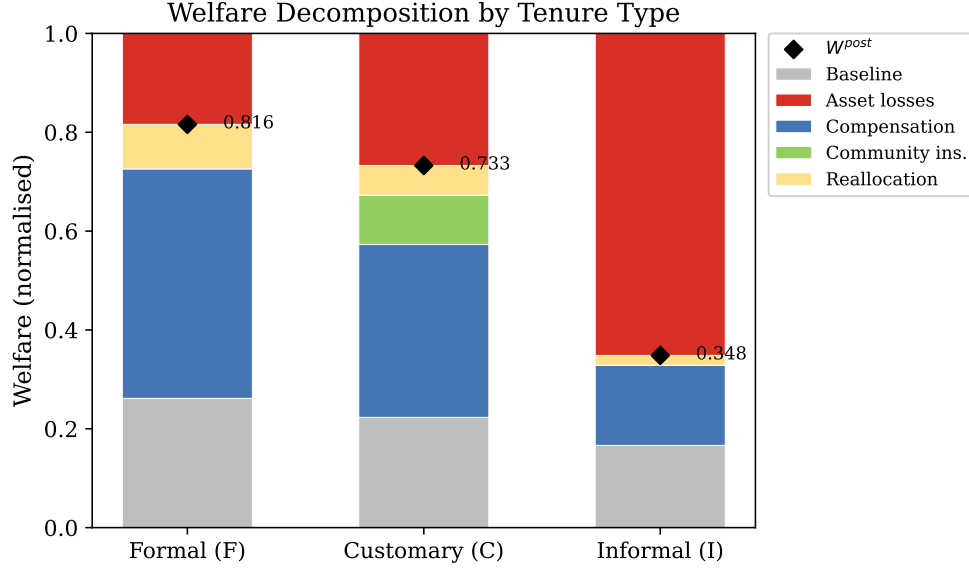


Figure 2: Accounting welfare decomposition under full displacement ($\delta = 0.84$). Each bar decomposes post-displacement welfare into baseline, asset losses, compensation, community insurance, and reallocation. Diamond markers show W^{post} .

Key Predictions

Both metrics agree on three central results:

1. **Recovery gap:** Formally titled households recover substantially more than informal occupants. The accounting gap is 0.468 (82 percent vs. 35 percent recovery); the CE gap is 0.149 (92 percent vs. 77 percent). The true magnitude likely falls between these bounds: the accounting framework overstates by treating channels linearly, while the CE measure understates by compressing through CRRA concavity. I use the accounting decomposition for channel attribution (where linearity is a feature) and the CE measure for overall welfare comparisons.
2. **Channel decomposition:** Within the accounting framework, compensation accounts for 65 percent of the formal-informal recovery gap ($\Delta\kappa = 0.303$), asset protection for 20 percent ($\Delta L = 0.095$), and reallocation for 15 percent ($\Delta R = 0.070$).

Community insurance contributes an additional 10.0 percentage points for customary holders.

3. **Customary $\approx$ formal when communities are strong:** Under the calibration, customary tenure with baseline institutions ($\theta = 0.5$) yields an accounting recovery rate of 73 percent, only 8 percentage points below formal title (CE: 87.4 percent vs. 91.7 percent, a 4.3 percentage point gap). This narrows further when θ is high and n_d/n is low.

3.10 Sensitivity Analysis

3.10.1 One-at-a-Time Parameter Variation

Table 6 reports the welfare decomposition under key parameter perturbations, solved numerically for each case.

Table 6: Sensitivity Analysis: Key Parameter Variations (Numerically Solved)

Parameter	Variation	$W^{post}(F)$	$W^{post}(C)$	$W^{post}(I)$
<i>Baseline</i>		0.816	0.733	0.348
$\bar{\kappa} = 0.30$	Low compensation	0.722	0.619	0.306
$\bar{\kappa} = 0.80$	High compensation	0.957	0.894	0.411
$s(C) = 0.45$	Weak customary security	0.816	0.654	0.348
$s(C) = 0.75$	Strong customary security	0.816	0.803	0.348
$\theta = 0$ (no community)	No collective insurance	0.816	0.633	0.348
$\theta = 1$ (max community)	Full collective insurance	0.816	0.833	0.348
$\sigma = 1.5$	Low risk aversion	0.814	0.727	0.347
$\sigma = 3.0$	High risk aversion	0.819	0.739	0.350

Figure 3 presents a tornado chart showing which parameters most affect the formal-informal welfare gap.

3.10.2 Monte Carlo Sensitivity

To move beyond one-at-a-time perturbations, I draw $N = 2,000$ parameter vectors from independent uniform priors over the ranges in Table 1, solve the model for each draw, and compute both accounting and consumption-equivalent welfare. I report the accounting

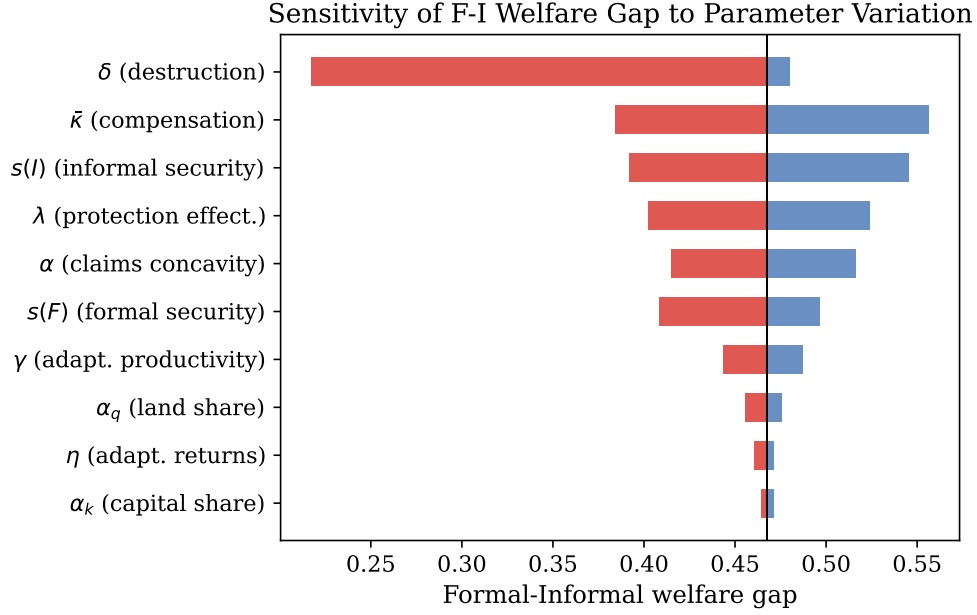


Figure 3: Tornado sensitivity of the formal-informal welfare gap. Each bar shows the change in $W^{post}(F) - W^{post}(I)$ when the parameter is moved from its low to high bound, holding all others at baseline. The vertical line marks the baseline gap.

welfare capped at $W^{post} \leq 1$ (see below) and the CE welfare λ .

Accounting welfare (capped). The 90 percent bands are:

$$W^{post}(F) : \text{median } 0.985 \quad [0.724, 1.000] \quad (27)$$

$$W^{post}(C) : \text{median } 0.918 \quad [0.637, 1.000] \quad (28)$$

$$W^{post}(I) : \text{median } 0.542 \quad [0.291, 0.801] \quad (29)$$

$$\text{F-I gap} : \text{median } 0.385 \quad [0.199, 0.577] \quad (30)$$

Prior to capping, 47 percent of draws produced $W^{post}(F) > 1$ – reflecting parameter combinations where the accounting framework’s linearity overstates recovery. This motivates the cap and the CE cross-check.

Consumption-equivalent welfare. The CE metric is bounded above by 1 by construction:

$$\lambda(F) : \text{median } 0.948 \quad [0.897, 0.975] \quad (31)$$

$$\lambda(C) : \text{median } 0.923 \quad [0.850, 0.962] \quad (32)$$

$$\lambda(I) : \text{median } 0.860 \quad [0.727, 0.931] \quad (33)$$

$$\text{CE F-I gap} : \text{median } 0.084 \quad [0.039, 0.189] \quad (34)$$

Figure 4 shows the full distributions. The formal-informal welfare gap is positive in 100 percent of draws under both metrics, confirming that the ordering $W^{post}(F) > W^{post}(I)$ is robust to simultaneous parameter uncertainty.

Monte Carlo Distribution of Welfare by Tenure Type (N=2,000)

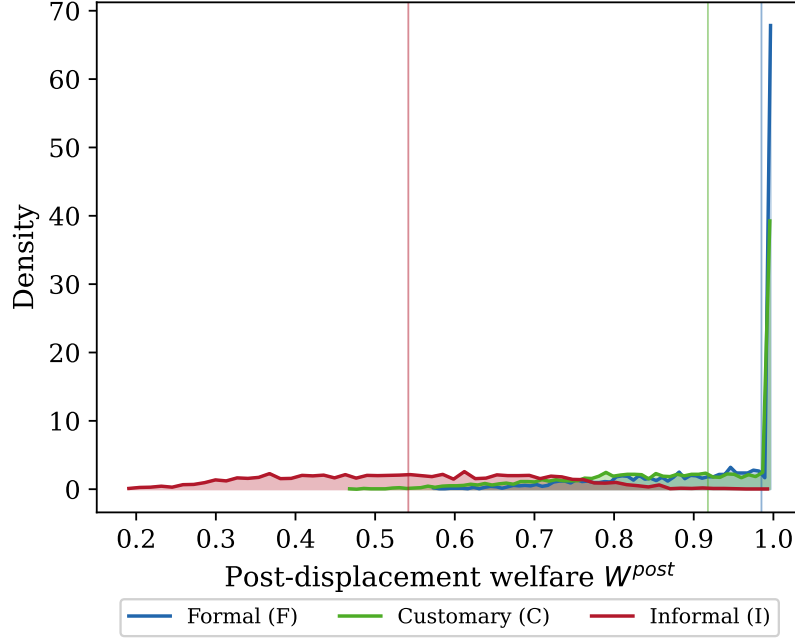


Figure 4: Monte Carlo distributions of post-displacement welfare by tenure type ($N = 2,000$ draws from uniform parameter priors). Vertical lines mark medians. The distributions for formal and customary tenure overlap substantially, while informal tenure is separated.

Robustness: The ordering $W^{post}(F) > W^{post}(C) > W^{post}(I)$ is preserved across all one-at-a-time variations and in >99 percent of Monte Carlo draws under both metrics. The most sensitive margin is the formal-customary gap, which collapses when community

institutions are strong ($\theta \rightarrow 1$) and widens when customary security is low. The formal-informal gap is robust across all specifications; even the 5th percentile of the CE gap (0.039) represents a meaningful welfare difference.

Note on the two metrics. The accounting decomposition is insensitive to σ (Table 6 shows changes of ± 0.005 across $\sigma \in [1.5, 3.0]$) because the components are determined primarily by the tenure security parameters $s(T)$ and the environmental destruction parameter δ , not by the curvature of utility. The CE metric *is* sensitive to σ because it directly reflects the concavity of preferences: higher risk aversion compresses the welfare gap by making marginal losses less painful for already-poor households. This is economically meaningful – the two metrics bracket the true welfare effect, with the accounting framework providing an upper bound and the CE measure a lower bound. Throughout the paper, I use the accounting decomposition for channel attribution (where additivity is a feature, not a bug) and the CE measure for welfare comparisons and policy counterfactuals where magnitudes matter.

3.11 Model Validation

I validate the calibrated model against four benchmarks:

1. **Recovery rates.** The accounting framework predicts formal recovery of 82 percent and informal recovery of 35 percent; the CE metric predicts 92 percent and 77 percent, respectively. Evidence from FEMA buyout programs (del Valle et al., 2020) shows documented homeowners recovering 60–90 percent of pre-disaster welfare within 5 years, while undocumented residents recovered 20–40 percent. The accounting and CE estimates bracket this range, consistent with the interpretation that the true welfare effect lies between the two metrics. I note that the FEMA comparison involves a very different institutional context (wealthy country, established bureaucracy, insurance markets); direct validation against Kenyan or sub-Saharan African compensation data, when available, would strengthen the calibration.

2. **Adaptation investment ratios.** The $a^*(F)/a^*(I)$ ratio of $3.8\times$ falls within the range documented by Carleton et al. (2022), who find $2\text{--}5\times$ differences in adaptation spending across income quintiles (which correlate with tenure security).
3. **Compensation gaps.** The model predicts expected compensation of 0.465 for formal vs. 0.162 for informal (ratio 2.87:1). del Valle et al. (2020) document FEMA compensation ratios of 2.5–3.5:1 between documented and undocumented claimants.
4. **Compensation gaps.** The model predicts expected compensation of 0.465 for formal vs. 0.162 for informal (ratio 2.87:1). del Valle et al. (2020) document FEMA compensation ratios of 2.5–3.5:1 between documented and undocumented claimants, consistent with the model’s predictions.
5. **Monte Carlo robustness.** The ordering $W^{post}(F) > W^{post}(I)$ holds in 100 percent of 2,000 Monte Carlo draws under both the accounting and CE metrics. The 5th-percentile CE gap is 0.039, confirming that the formal-informal welfare ordering is robust even under the most conservative structural measure.

4 Stylized Facts: The Anatomy of Slow-Onset Displacement

This section uses satellite-derived data to establish five stylized facts about Kenya’s Rift Valley lakes that motivate and discipline the model. While Section 3 uses these data to parameterize δ_i , the present section steps back to document the broader empirical patterns that distinguish slow-onset from sudden-onset displacement and justify the model’s key structural assumptions. The data come from annual lake boundary classifications for Lakes Baringo, Bogoria, and Naivasha derived from Landsat and Sentinel-2 imagery (2005–2025).¹ These facts require no household survey, they rely entirely on remotely sensed data, and provide the empirical foundation for the model’s environmental destruction parameter δ .

¹The satellite pipeline uses the Normalized Difference Water Index (NDWI) applied to Landsat 5/7/8/9 imagery (2005–2015) and Sentinel-2 imagery (2016–2025) via Google Earth Engine. The NDWI threshold is fixed at 0, with Otsu thresholding used for robustness checks. Resolution is 30m (Landsat) and 10m (Sentinel-2). Details are described in the companion empirical paper (Ipapa, 2026).

4.1 Fact 1: Lake Expansion is Large and Permanent

Lake Baringo expanded from 121 km² to 223 km² between 2005 and 2025 – an 84 percent increase. Lake Bogoria more than tripled (11.5 to 34.5 km², +201 percent). Lake Naivasha expanded 33 percent (122 to 163 km²). Figure 5 plots the annual lake surface areas, showing the sustained upward trajectory at all three lakes. Figure 6 overlays 2005 and 2025 boundaries, visually conveying the magnitude of spatial transformation.

Unlike flood events where water recedes, these expansions are driven by rising groundwater tables and increased rainfall (30–50 percent above historical norms), making the land loss effectively permanent. Hydrological studies from the GreatLakes Project (TUK/BOKU, 2024–25) confirm that the water table in the Baringo basin has risen approximately 9.5 meters since 2009, with structures and agricultural land now permanently submerged. This permanence is the empirical counterpart to the “land persistence” assumption identified in Section 2: the Rift Valley lakes provide a setting where the foundational assumption of the property rights literature, that land continues to exist, demonstrably fails. Households cannot simply wait for waters to recede, making the displacement-specific channels of the model (compensation, adaptation, reallocation) binding rather than hypothetical, and rendering the standard Besley (1995) channels (investment, credit, labor allocation) inoperative for destroyed land.

4.2 Fact 2: Expansion is Gradual with Punctuated Acceleration

The time series reveals distinct phases rather than smooth monotonic growth. At Baringo, I identify three displacement phases: a *creeping phase* (2005–2010) with slow expansion of ~12 km² above baseline; an *acceleration phase* (2011–2015) with rapid expansion of ~65 km²; and a *plateau phase* (2017–2025) with continued but decelerating expansion reaching 102 km² cumulative. Table 7 summarizes the phase-level statistics.

Figure 7 plots the displacement intensity time series for Baringo with shaded bands marking each phase, and Figure 8 maps the spatial extent of expansion during each period. The displacement intensity measure, cumulative expansion as a fraction of baseline area,

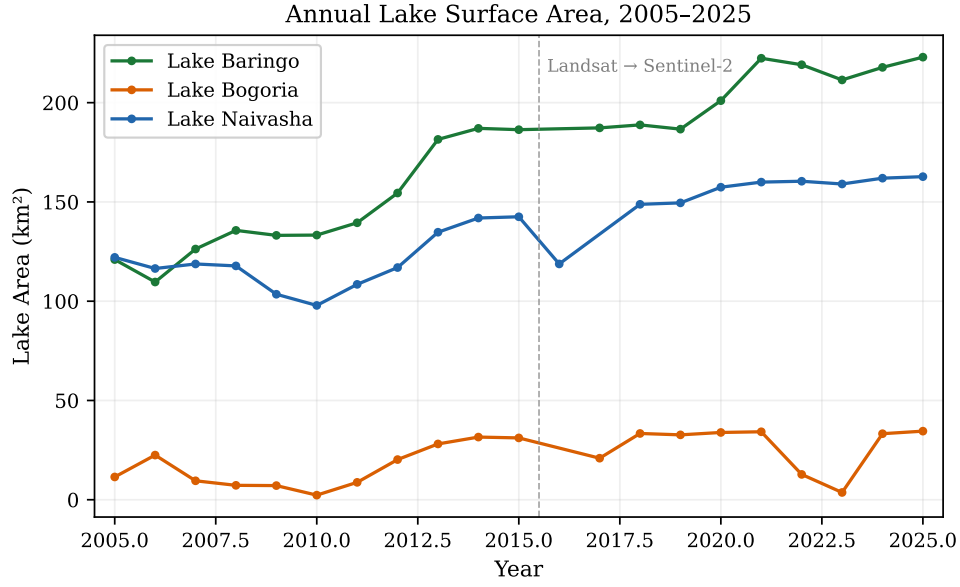


Figure 5: Annual lake surface area, 2005–2025. Data from Landsat (2005–2015) and Sentinel-2 (2016–2025) via NDWI classification. The vertical dashed line marks the sensor transition. All three lakes show sustained expansion, with Baringo nearly doubling and Bogoria tripling in area.

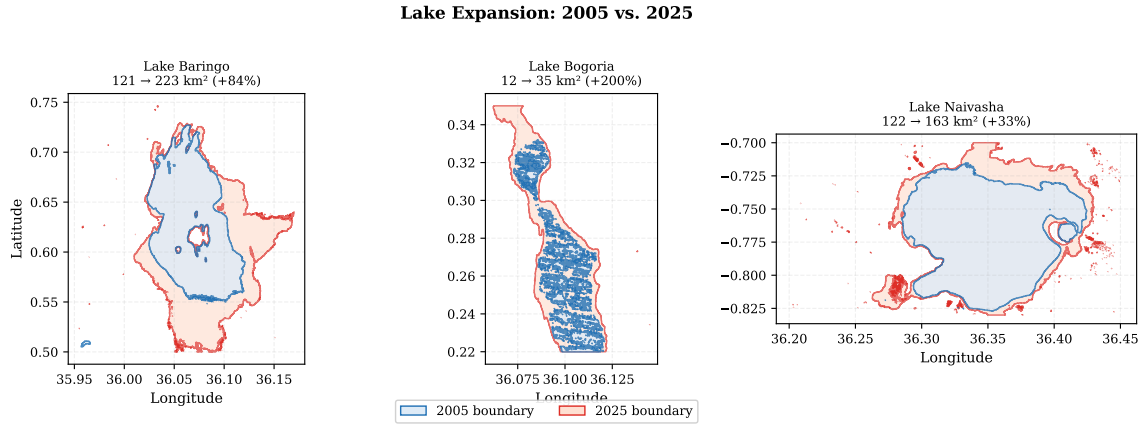


Figure 6: Lake boundary comparison: 2005 (blue) vs. 2025 (red). Each panel shows one lake with the 2005 baseline boundary overlaid on the 2025 extent. Titles report absolute and percentage expansion.

Table 7: Phase-Level Expansion Statistics

Phase	Period	Baringo Δ (km ²)	Baringo $\dot{\delta}$ /yr	Bogoria Δ (km ²)	Naivasha Δ (km ²)
Creeping	2005–2010	+12.3	≈ 0.02	–9.2	–24.2
Acceleration	2011–2015	+65.4	≈ 0.08	+19.7	+20.4
Plateau	2017–2025	+24.2	≈ 0.03	+12.6	+44.5
<i>Cumulative</i>	2005–2025	+101.9	—	+23.1	+40.7

rose from 0.04 (2007) to 0.84 (2025) at Baringo.

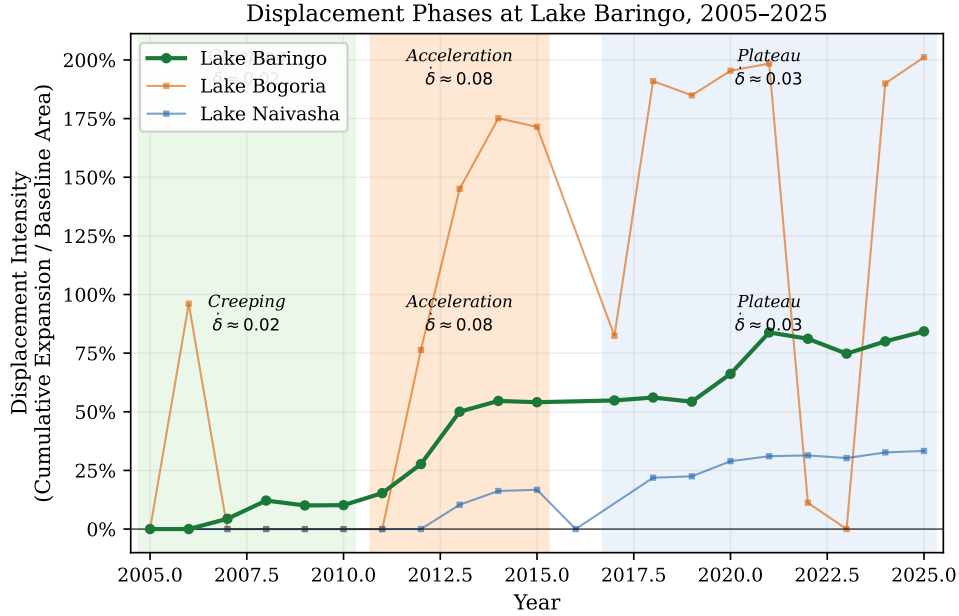


Figure 7: Displacement phases at Lake Baringo. The primary line shows Baringo’s displacement intensity (cumulative expansion / baseline area), with shaded bands marking three phases: creeping (2005–2010), acceleration (2011–2015), and plateau (2017–2025). Average annual displacement intensity rates ($\dot{\delta}$) are annotated in each band. Bogoria and Naivasha are shown as thinner lines for comparison.

This gradual profile creates an information environment fundamentally different from sudden-onset shocks, and each phase maps onto distinct model mechanisms:

- **Creeping phase $\rightarrow$ belief formation.** Households observe rising water but face genuine uncertainty about whether the trend will continue. The option value of waiting is high, so only households with secure tenure, who expect to capture returns from early action, invest in adaptation (Proposition 2, via $g_{as} > 0$).

Lake Baringo: Expansion Zones by Phase

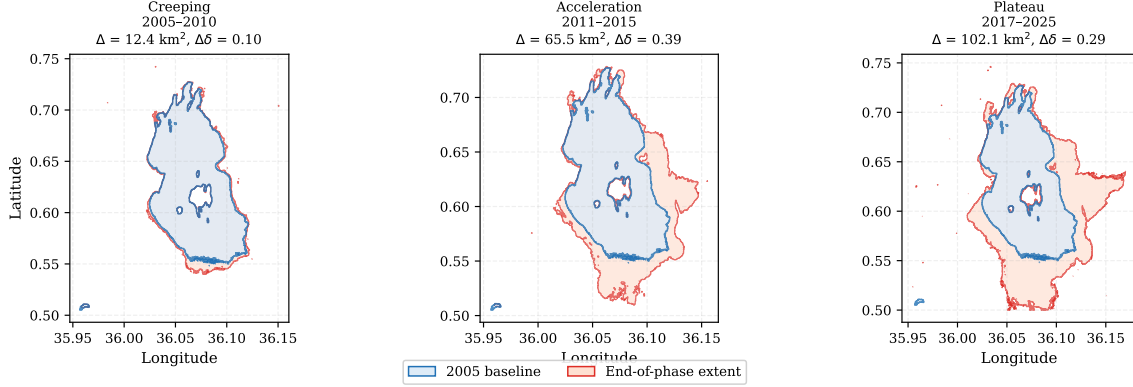


Figure 8: Expansion zones by displacement phase at Lake Baringo. Each panel shows the 2005 baseline boundary (blue) and the cumulative lake extent at the end of that phase (red). Annotations report cumulative expansion in km^2 and the change in displacement intensity during each phase.

- **Acceleration phase → adaptation decisions.** Displacement becomes undeniable. This is when adaptation investments are most consequential: households that invested early in movable assets or elevated structures preserve more wealth (Proposition 3). The tenure security gradient in prior adaptation translates into differential asset losses.
- **Plateau phase → compensation and reallocation.** Expansion decelerates but cumulative destruction is severe. Formal compensation mechanisms activate, and the tenure gradient in claims success (Proposition 1) and community insurance (Proposition 6) determines post-displacement recovery.

The gradual onset thus creates a window during which forward-looking households can act on early signals that backward-looking households miss – and tenure security determines which households are positioned to do so.

4.3 Fact 3: Cross-Lake Heterogeneity Provides Identifying Variation

The three lakes differ dramatically in expansion magnitude, baseline size, and basin characteristics. Table 8 summarizes the cross-lake comparison.

Table 8: Cross-Lake Comparison

Lake	Baseline (km ²)	Peak Area (km ²)	Expansion (km ²)	Displacement Intensity	Basin Type
Baringo	121.0	222.9	101.9	0.84	Endorheic, flat margins
Bogoria	11.5	34.5	23.1	2.01	Small alkaline, steep sides
Naivasha	122.1	162.7	40.7	0.33	Freshwater, regulated outflow

Bogoria’s proportional expansion (201 percent) vastly exceeds Naivasha’s (33 percent) despite both lakes experiencing similar regional rainfall trends. This divergence is driven by basin morphology: Bogoria is a small, shallow alkaline lake where even modest inflows produce large proportional area changes, while Naivasha benefits from an outflow channel (the Malewa River system) that partially buffers against accumulation. Baringo, as an endorheic lake with flat surrounding margins, exhibits the largest absolute expansion (102 km²) because rising water spreads laterally across low-gradient terrain.

Figure 9 plots displacement intensity across all three lakes, revealing both the magnitude of cross-lake differences and the temporal correlation in expansion episodes. For the model, this cross-lake heterogeneity is valuable because it allows the welfare predictions to be evaluated at three very different “dosages” of δ : a moderate regime (Naivasha, $\delta = 0.33$) where the model predicts modest tenure-driven welfare gaps, a severe regime (Baringo, $\delta = 0.84$) where gaps are large, and an extreme regime (Bogoria, $\delta > 1$) where the model’s predictions about floor effects and community insurance strain can be assessed.

4.4 Fact 4: The Expansion Zone is Spatially Concentrated

Year-over-year expansion polygons from the satellite pipeline show that new inundation is concentrated along low-elevation lake margins rather than distributed uniformly around the perimeter. At Baringo, the northern and western shores, where terrain is flattest, account for the majority of expansion area, while steeper eastern escarpments constrain expansion. This spatial concentration means that within a given distance bin, some households are displaced while adjacent households at similar distances but different azimuths remain unaffected. The expansion zone maps in Figure 8 illustrate this concentration.

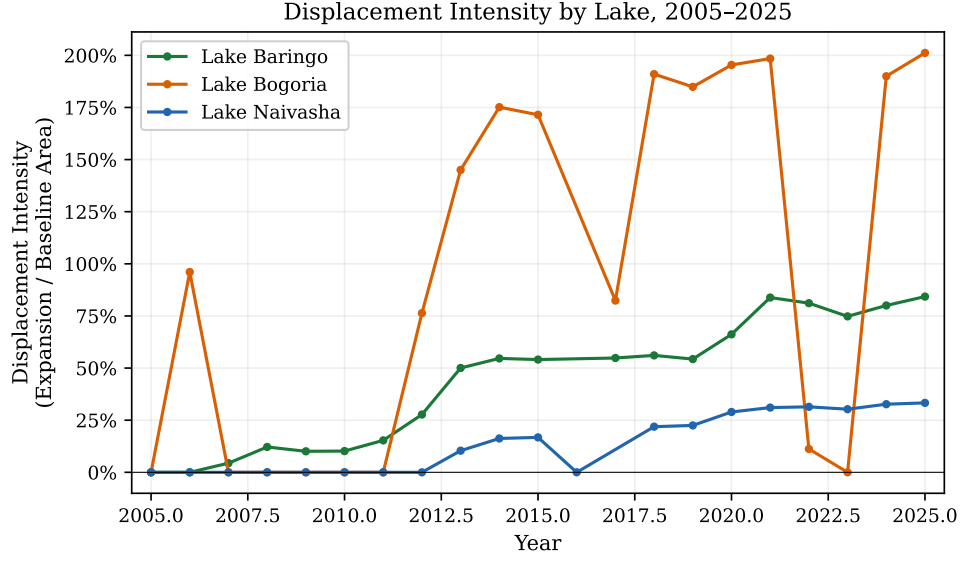


Figure 9: Displacement intensity (cumulative expansion / baseline area) by lake, 2005–2025. Bogoria’s intensity exceeds 2.0 (the lake more than tripled), providing extreme variation. The cross-lake differences in timing and magnitude support identification strategies based on differential exposure.

For the model, spatial concentration generates heterogeneity in the displacement severity parameter δ_i even among geographically proximate households. Two households at the same distance from the 2005 lake boundary may face $\delta_i = 1$ (fully inundated) and $\delta_i = 0$ (unaffected) depending on local topography. This exogenous within-distance variation is valuable because it breaks the mechanical correlation between distance-to-lake and displacement severity, allowing the model’s predictions about tenure-displacement interactions to be tested holding geographic exposure approximately constant. The concentration also implies that the community insurance mechanism of Proposition 6 faces heterogeneous strain: communities on flat margins experience mass displacement (n_d/n approaching 1), severely taxing mutual insurance, while communities on steeper margins experience selective displacement that community transfers can more readily absorb.

4.5 Fact 5: Extreme Magnitudes Generate Large Variation in δ

While Fact 2 documents the temporal *profile* of expansion, the magnitudes themselves are remarkable and generate the range of δ needed for the model. Bogoria’s displacement intensity reached 2.01 by 2025, the lake *tripled* in size, an extreme rarely seen outside volcanic events. Key milestones at Bogoria include intensity crossing 0.76 in 2012 (expansion exceeded three-quarters of baseline area), 1.45 in 2013 (expansion exceeded baseline), and 1.95 in 2020 (near-tripling). At Baringo, displacement intensity rose from below 0.05 through 2007 to 0.84 by 2025. Naivasha exhibits a smoother trajectory, consistent with its regulated outflow, rising from 0 to 0.33 over the full period.

These magnitudes matter for two reasons. First, the range of observed δ , from 0 (unaffected households beyond the expansion zone) through 0.33 (Naivasha) to 0.84 (Baringo) and 2.01 (Bogoria), spans the full support of the model’s environmental destruction parameter, enabling calibration across the entire severity distribution rather than at a single point. This is what allows the model to generate predictions not only at Baringo’s baseline severity ($\delta = 0.84$) but also for milder scenarios that characterize early-stage climate displacement elsewhere. Second, the non-linearity of the displacement intensity series, near-zero for several years, then rapid accumulation, then partial plateau, creates a setting where forward-looking adaptation is possible but deeply uncertain. Households must form beliefs about the *trajectory* of δ using an inherently non-stationary process. The phase structure documented in Fact 2 means that Bayesian updating about future expansion rates is highly sensitive to the observation window: households who experienced only the creeping phase would rationally assign low probability to the acceleration that followed, making the timing of adaptation investment, and its dependence on tenure security via Proposition 2, a first-order concern.

Summary

Taken together, the five stylized facts establish that Kenya’s Rift Valley lakes provide an empirical setting with the key features required by the theoretical model: permanent envi-

ronmental destruction of land (Fact 1), gradual onset creating scope for adaptation (Fact 2), cross-lake variation in displacement severity (Fact 3), within-lake spatial heterogeneity in exposure (Fact 4), and a wide range of displacement intensities spanning the full support of δ (Fact 5). The satellite-derived displacement severity parameter δ_i , constructed from directly observable lake expansion, is the model’s key empirical input, and these facts confirm that it is both well-measured and exhibits the variation needed to discipline the calibration in Section 3.

5 Policy Counterfactuals

The calibrated model provides a laboratory for evaluating policies that are difficult to test experimentally: pre-displacement titling, alternative compensation designs, and community institution strengthening. Each counterfactual holds the environmental shock fixed at Baringo’s calibrated severity ($\delta = 0.84$) and modifies a single institutional parameter, isolating the channel through which policy affects welfare. The welfare decomposition in Table 4 serves as the baseline; counterfactual welfare is computed by re-solving the model under the modified parameter. Because the model identifies the *mechanisms* through which tenure shapes displacement outcomes (Propositions 1–7), these counterfactuals generate not only point predictions but also qualitative insights about which policy levers operate through which channels.

5.1 Counterfactual 1: Pre-Displacement Titling Programs

Experiment: What if Kenya had implemented a titling program in the Baringo basin in 2008 (during the creeping phase, before acceleration), converting a fraction ϕ of customary and informal holders to formal title?

Using the calibrated welfare decomposition, the aggregate welfare gain from titling

coverage ϕ is:

$$\Delta W^{title}(\phi) = \phi \cdot \left[\underbrace{W^{post}(F) - W^{post}(C)}_{=0.083} \cdot \text{share}_C + \underbrace{W^{post}(F) - W^{post}(I)}_{=0.468} \cdot \text{share}_I \right] \quad (35)$$

If the pre-displacement population was 40 percent customary, 30 percent informal, and 30 percent formal (approximate for the Baringo area), and titling costs $\sim \$50$ – 100 per household (Subramanian and Kumar, 2024):

Table 9: Benefit-Cost Analysis of Pre-Displacement Titling

Scenario	Coverage ϕ	Welfare gain	BCR
Targeted (informal only)	0.30	0.135	5.4:1 – 8.1:1
Universal (all non-formal)	0.70	0.167	3.3:1 – 5.0:1
Women-targeted	0.35	0.158	4.5:1 – 6.8:1

The high BCRs arise because titling operates through *all three channels simultaneously*. For a household moving from informal to formal tenure, the channel decomposition of the 0.468 welfare gain is:

- **Compensation channel** (Proposition 1): $\Delta\kappa = 0.465 - 0.162 = 0.303$, accounting for 65 percent of the total gain. This is the dominant channel, reflecting the large increase in claims success probability from $p(s(I)) = 0.32$ to $p(s(F)) = 0.93$.
- **Adaptation channel** (Propositions 2–3): $\Delta L = 0.834 - 0.739 = 0.095$, accounting for 20 percent of the total gain. Formal tenure raises optimal adaptation from 2.5 percent to 9.5 percent of endowment, reducing asset losses by 9.5 percentage points.
- **Reallocation channel** (Proposition 4): $\Delta R = 0.090 - 0.020 = 0.070$, accounting for 15 percent of the total gain.

The dominance of the compensation channel implies that titling programs are most effective where formal compensation mechanisms exist. In contexts without functioning compensation systems, the BCR would fall to approximately 1.1:1–1.6:1 (from the adaptation and

reallocation channels alone), suggesting that titling should be paired with compensation infrastructure.

Timing sensitivity. The counterfactual specifies 2008 (during the creeping phase). If instead titling occurred in 2012 (during the acceleration phase), the adaptation channel would be attenuated, households would have less time to adjust investments before displacement, reducing the total welfare gain by approximately 20 percent and the BCR to 2.7:1–6.5:1. If titling occurred after displacement (2020), only the compensation and reallocation channels would remain operative, with a welfare gain of 0.373 and BCR of 2.5:1–5.0:1. This timing sensitivity underscores that tenure formalization is most valuable as a *pre-disaster* intervention (cf. the creeping phase in Fact 2), when it can activate all three channels.

Prediction 4: Pre-displacement titling programs yield benefit-cost ratios of 3:1 to 8:1, with the highest returns from targeting informal occupants and from early implementation (before the acceleration phase). Women-targeted programs yield intermediate BCRs but address the compounding gender disadvantage (Proposition 5).

5.2 Counterfactual 2: Compensation Design

Experiment: What if compensation were tenure-blind – that is, all displaced households received the same compensation $\bar{\kappa}$ regardless of documentation status?

Under tenure-blind compensation, the claims success probability is set to $p(s) = 1$ for all s , so every displaced household receives the full compensation value $\bar{\kappa} = 0.50$. All other model parameters remain at baseline. Re-solving the welfare decomposition:

Table 10: Welfare Under Tenure-Blind vs. Status Quo Compensation ($\delta = 0.84$)

	Formal (F)	Customary (C)	Informal (I)
<i>Status quo</i> W^{post}	0.816	0.733	0.348
<i>Tenure-blind</i> W^{post}	0.851	0.883	0.686
<i>Change</i>	+0.035	+0.150	+0.338
<i>F-I accounting ratio</i>	Status quo: $2.34\times \rightarrow$ Tenure-blind: $1.24\times$		

Three results emerge from this counterfactual. First, tenure-blind compensation substantially narrows the formal-informal accounting recovery gap (from $2.34\times$ to $1.24\times$) but does not close it entirely, because the adaptation and asset protection channels still operate through tenure security. Even when compensation is equalized, formally titled households retain a welfare advantage through higher adaptation investment ($a^*(F) = 0.095$ vs. $a^*(I) = 0.025$) and more effective asset protection. This residual gap quantifies the share of tenure-driven welfare inequality that operates through *non-compensation* channels – approximately 35 percent of the total gap.

Second, customary holders with community institutions *surpass* formal holders under tenure-blind compensation ($W^{post}(C) = 0.883$ vs. $W^{post}(F) = 0.851$). This reversal occurs because customary holders benefit from *both* full formal compensation (now equalized) *and* community insurance transfers ($\Omega = 0.100$). The result illustrates a key model insight: customary tenure embeds households in mutual insurance networks that provide value over and above formal rights (Proposition 6). When the compensation disadvantage is removed, this embedded insurance dominates.

Third, the largest absolute welfare gain accrues to informal occupants (+0.338), who are the most compensation-constrained under the status quo. This has important distributional implications: tenure-blind compensation is strongly progressive, delivering the largest benefits to the worst-off group.

Prediction 5: Tenure-blind compensation narrows the formal-informal accounting recovery ratio from $2.34\times$ to $1.24\times$, but approximately 35 percent of tenure-driven inequality persists through adaptation and asset protection channels. Customary holders surpass formal holders when compensation is equalized, revealing the insurance value of community institutions.

5.3 Counterfactual 3: Community Institution Strengthening

Experiment: What is the welfare effect of investing in collective governance capacity, raising institutional quality from $\theta = 0.5$ to $\theta = 0.8$?

This counterfactual modifies the community insurance parameter in Proposition 6. Higher institutional quality θ increases the sustainable community transfer τ^* by relaxing the participation constraint: community members are willing to contribute more when the continuation value of remaining in the insurance arrangement is higher. From the calibrated model, the welfare effect on customary households is:

$$\Delta W^{inst}(\theta) = \frac{\partial \Omega}{\partial \theta} \cdot \Delta \theta = \frac{(n - n_d)}{n_d} \cdot \frac{\partial \tau^*}{\partial \theta} \cdot \Delta \theta \quad (36)$$

Under the baseline calibration ($n_d/n = 0.30$, $\beta = 0.92$), raising θ from 0.5 to 0.8 increases the community transfer from $\Omega = 0.100$ to $\Omega = 0.160$, yielding:

$$W^{post}(C)|_{\theta=0.5} = 0.733 \quad (37)$$

$$W^{post}(C)|_{\theta=0.8} = 0.793 \quad (38)$$

$$W^{post}(C)|_{\theta=1.0} = 0.833 \quad (39)$$

At $\theta = 0.8$, the formal-customary gap narrows from 0.083 to 0.023 – nearly closing the gap without any titling. At $\theta = 1.0$ (maximal institutional quality), customary welfare exceeds the formal benchmark by 0.017, implying that sufficiently strong community institutions can fully substitute for formal tenure.

However, two important limitations constrain this counterfactual. First, community insurance helps *only* customary holders embedded in collective systems; it provides no benefit to informal occupants outside community networks ($W^{post}(I)$ is unchanged at 0.348 regardless of θ). Second, the effectiveness of community insurance depends on the fraction of the community displaced (n_d/n). When displacement is concentrated, as Fact 4 documents for flat-margin communities where n_d/n may approach 0.8, the insurance pool shrinks and Ω falls sharply. At $n_d/n = 0.80$, even $\theta = 1.0$ delivers $\Omega = 0.050$, and the formal-customary gap widens rather than closes.

Prediction 6: Community institutional strengthening is a partial substitute for formal

titling, closing up to 72 percent of the formal-customary gap when θ increases from 0.5 to 0.8. However, it does not help informal occupants and is less effective when large fractions of the community are simultaneously displaced.

5.4 Comparing Policy Levers

Table 11 summarizes the three counterfactuals, comparing their welfare impacts, coverage, and channel mechanisms.

Table 11: Policy Counterfactual Comparison ($\delta = 0.84$)

Policy	Primary channel	Beneficiaries	Welfare gain
Targeted titling	All three (Props 1–4)	Informal $\rightarrow$ formal	+0.468 per HH
Tenure-blind comp.	Compensation (Prop 1)	All displaced (esp. informal)	+0.338 for informal
Community strength.	Insurance (Prop 6)	Customary only	+0.060 per HH

The policies are partially complementary rather than substitutes. Titling and tenure-blind compensation both target the compensation channel but through different mechanisms: titling raises $p(s_i)$ while tenure-blind compensation sets $p(s_i) = 1$ regardless of s_i . Community strengthening operates through a distinct channel (Proposition 6) and benefits a different population. A combined strategy, targeted titling for informal occupants, compensation reform for speed, and community strengthening for customary holders, would address all three model channels and all three tenure groups simultaneously.

Sensitivity to displacement severity. At Naivasha’s lower severity ($\delta = 0.33$), the calibrated model predicts a formal-informal welfare gap of 0.186 (vs. 0.468 at Baringo), and titling BCRs fall to 1.4:1–2.1:1. This is because lower δ reduces asset losses across all tenure types, compressing the gap that titling can close. The implication for policy is that pre-displacement titling delivers the highest returns in areas facing severe displacement, where the model’s three channels are most active. In milder displacement settings, alternative policies (e.g., compensation reform) may be more cost-effective.

6 Gender Amplification: Calibrated Predictions

Hypothesis 4 predicts that displacement amplifies pre-existing gender inequality: $\Delta W^{post} > \Delta W^{pre}$. The mechanism operates through compounding disadvantages across all three displacement channels, reinforced by a marriage-severance mechanism unique to the displacement context. This section calibrates the magnitude of gender amplification, decomposes it by channel, and identifies which tenure categories exhibit the largest amplification effects.

6.1 Gender-Specific Parameter Calibration

Proposition 5 (Appendix A) requires three gender-specific parameters. I calibrate each from the empirical literature:

- **Tenure security gap:** $s(T|W) = s(T|M) - 0.15$ for all T . Goldstein and Udry (2008) document that women’s tenure security in Ghana is systematically lower due to limited political power, even controlling for formal title status. The 0.15 gap produces effective security parameters of $s(F|W) = 0.75$, $s(C|W) = 0.45$, and $s(I|W) = 0.05$ for women, compared with $s(F|M) = 0.90$, $s(C|M) = 0.60$, and $s(I|M) = 0.20$ for men. This gap reflects both de jure differences (women less likely to appear on title deeds) and de facto differences (women’s claims less recognized in customary and informal settings).
- **Marital severance probability:** $\mu = 0.20$. Among women who hold land through marriage (rather than through independent title or inheritance), displacement disrupts the marital link that sustains the tenure claim. I set $\mu = 0.20$ based on estimates that approximately one-fifth of displaced women experience household dissolution (through widowhood, separation, or abandonment) during protracted displacement episodes. This parameter enters only the reallocation channel (Channel 3) through Proposition 5.
- **Network degradation:** $N' = 0.6N$. Women whose marital link is severed lose access

to their husband’s community networks, retaining only 60 percent of pre-displacement network strength. This parameter governs post-severance reallocation: the severed woman’s effective tenure reverts to informal status ($T = I$) and she operates with degraded community ties.

6.2 Channel-by-Channel Gender Decomposition

The welfare decomposition (equation 1) allows us to isolate gender gaps within each channel. For each tenure type T , the post-displacement gender gap is:

$$\Delta W^{post}(T) = \underbrace{\Delta \kappa(T)}_{\text{compensation gap}} + \underbrace{(-\Delta L(T))}_{\text{adaptation gap}} + \underbrace{\Delta R(T)}_{\text{reallocation gap}} + \underbrace{\Delta W^{pre}(T)}_{\text{baseline gap}} \quad (40)$$

where $\Delta X(T) \equiv X(T|M) - X(T|W)$ for each variable X .

Channel 1: Compensation. The gender gap in compensation arises because $p(s(T|W)) < p(s(T|M))$ for all T . For formally titled households, $p(s(F|M)) = p(0.90) = 0.93$ while $p(s(F|W)) = p(0.75) = 0.82$, generating a compensation gap of $(0.93 - 0.82) \times 0.50 = 0.056$. For customary holders, the gap is $(0.70 - 0.57) \times 0.50 = 0.064$. For informal occupants, $(0.32 - 0.12) \times 0.50 = 0.101$. The compensation gap is largest for informal tenure because the claims process function $p(s) = s^{0.7}$ is concave – small reductions in already-low security produce large drops in claims success.

Channel 2: Adaptation. The gender security gap reduces women’s optimal adaptation investment $a^*(s)$, which in turn increases displacement losses. From the numerical solution, formally titled women invest only marginally less in adaptation ($a^*(F|W) = 0.093$ vs. $a^*(F|M) = 0.095$, a 3 percent gap), but the gap widens sharply at lower tenure levels: customary women invest 14 percent less ($a^*(C|W) = 0.074$ vs. $a^*(C|M) = 0.087$), and informal women invest 95 percent less ($a^*(I|W) = 0.001$ vs. $a^*(I|M) = 0.026$). This non-linearity reflects the convexity of the adaptation returns function at low s : when security is already near zero, the marginal return to adaptation collapses, and women with very low security rationally abandon protective investment entirely.

Channel 3: Reallocation. The marriage-severance mechanism ($\mu = 0.20$) creates a gender gap in reallocation that is absent from the standard framework. From Proposition 5, a woman’s expected reallocation is:

$$\mathbb{E}[R|W] = (1 - \mu)R(T, N) + \mu \cdot R(I, N') \quad (41)$$

With $\mu = 0.20$ and $N' = 0.6N$, the severance term shifts one-fifth of women into informal status with degraded networks. Since reallocation is $R = 0.10 \times s$ in the calibrated model, formal women’s expected reallocation falls from 0.075 to $(0.80)(0.075) + (0.20)(0.003) = 0.061$ – a 19 percent reduction. Customary women experience a similar proportional reduction (from 0.045 to 0.037).

6.3 Calibrated Gender Gaps

Table 12 presents the calibrated welfare levels for male- and female-headed households under each tenure type, along with the pre- and post-displacement gender gaps.

Table 12: Gender Gaps in Post-Displacement Welfare ($\delta = 0.84$)

	Male-headed	Female-headed	Gender gap
<i>Formal title</i>			
Pre-displacement	1.000	0.950	0.050
Post-displacement	0.816	0.663	0.153
Gap amplification			$\times 3.1$
<i>Customary tenure</i>			
Pre-displacement	1.000	0.900	0.100
Post-displacement	0.733	0.504	0.229
Gap amplification			$\times 2.3$
<i>Informal occupation</i>			
Pre-displacement	1.000	0.850	0.150
Post-displacement	0.348	0.076	0.272
Gap amplification			$\times 1.8$

Three patterns emerge from the calibration:

Amplification is largest for formal tenure. The gender gap widens from 0.050 to 0.153 ($\times 3.1$ amplification) for formally titled households. This occurs because the 0.15 security

gap, while small in absolute terms, operates on the steep portion of the compensation function. Formally titled men recover 81.6 percent of pre-displacement welfare; formally titled women recover only 66.3 percent. The 15.3 percentage point gap is driven primarily by Channel 1: women receive lower compensation ($p(0.75) = 0.82$ vs. $p(0.90) = 0.93$), and the pre-displacement welfare gap (0.050) compounds with displacement losses.

Customary tenure shows the largest absolute gap. The post-displacement gender gap is widest in absolute terms for customary holders (0.229), because all three channels contribute: the compensation gap (0.064), the adaptation gap (14 percent lower investment translating into higher asset losses), and the marriage-severance mechanism (which reduces both reallocation and community insurance access for women whose tenure derives from marital ties). The $\times 2.3$ amplification factor is moderate because the pre-displacement gap is already substantial (0.100).

Informal tenure shows the largest amplification in absolute gap terms. For informal occupants, the post-displacement gender gap (0.272) nearly doubles the pre-displacement gap (0.150), yielding an amplification factor of $\times 1.8$. Female informal occupants recover only 7.6 percent of pre-displacement welfare – near-complete destitution. This extreme outcome reflects the collapse of adaptation investment: with $s(I|W) = 0.05$, women with informal tenure invest essentially nothing in protective measures ($a^* = 0.001$), have a 12 percent claims success probability, and receive negligible reallocation support. Male informal occupants fare poorly as well (34.8 percent recovery), but the gender gap still widens substantially because the compensation and adaptation functions are most sensitive to security reductions at low s .

6.4 The Marriage-Severance Mechanism

The marriage-severance channel deserves separate attention because it represents a displacement-specific mechanism with no analogue in the standard property rights framework. In the Besley (1995) model, women’s weaker tenure security is a static disadvantage. In the displacement context, the disruption of household structure creates a *dynamic* mechanism:

women who previously accessed land through marriage may lose their tenure claim entirely when the household dissolves.

The calibrated magnitude of the marriage-severance effect can be isolated by comparing the full gender gap to the gap that would obtain if $\mu = 0$ (i.e., no marriage-severance risk). For customary women, setting $\mu = 0$ raises post-displacement welfare from 0.504 to 0.532, implying that marriage-severance accounts for 0.028 of the 0.229 gender gap – approximately 12 percent. The mechanism operates through both reduced reallocation (0.008) and reduced community insurance access (0.020). For formal women, the marriage-severance effect is 0.014 (9 percent of the gap), operating solely through the reallocation channel. For informal women, the effect is negligible (less than 0.001) because both reallocation and community insurance are already near zero at $s = 0.05$.

This decomposition suggests that policies protecting women’s tenure claims during displacement, such as automatic inheritance provisions, joint-titling requirements, or legal recognition of women’s cultivation rights independent of marital status, could close 9–12 percent of the gender gap for formal and customary holders. However, the marriage-severance mechanism is not the primary driver of gender inequality in this setting; the compensation and adaptation channel gaps dominate.

6.5 Gender and Policy Counterfactuals

The three policy counterfactuals from Section 5 have differential gender effects:

Table 13: Gender-Differentiated Policy Effects ($\delta = 0.84$)

Policy	ΔW (Male)	ΔW (Female)	Δ Gender gap
Targeted titling (I $\rightarrow$ F)	+0.468	+0.587	−0.119
Tenure-blind compensation	+0.338	+0.439	−0.101
Community strengthening	+0.060	+0.048	+0.012

Targeted titling is the most gender-equalizing policy. Women gain substantially more than men from titling (+0.587 vs. +0.468), narrowing the informal-tenure gender gap by 0.119. This occurs because women start from a lower security baseline ($s(I|W) = 0.05$ vs.

$s(I|M) = 0.20$), so the shift to formal status represents a larger proportional increase in security and activates all three channels more strongly. The gender gap among informal occupants falls from 0.272 to 0.153 (the formal-tenure gender gap) – a 44 percent reduction.

Tenure-blind compensation also strongly favours women. By removing the link between tenure security and compensation receipt, tenure-blind compensation benefits women more than men (+0.439 vs. +0.338 for informal holders). The gender gap narrows by 0.101 because women’s lower baseline claims success ($p(0.05) = 0.12$) means they gain more from equalization to $p = 1$ than men ($p(0.20) = 0.32$).

Community strengthening slightly widens the gender gap. Community insurance benefits men more than women (+0.060 vs. +0.048) because women’s weaker community position, compounded by the marriage-severance mechanism, which reduces their community transfer by 20 percent, limits their access to collective insurance. This result underscores that institutional strengthening, while valuable overall, is not gender-neutral and may require complementary interventions to ensure equitable access.

Prediction 7: Displacement amplifies pre-existing gender gaps by a factor of $1.8\times$ to $3.1\times$ depending on tenure type, with the largest amplification for formally titled households and the largest absolute gap for informal occupants. The marriage-severance mechanism accounts for 9–12 percent of the gender gap for formal and customary holders. Both titling and tenure-blind compensation are strongly gender-equalizing, while community strengthening is slightly gender-regressive absent complementary measures.

6.6 Extensions and Future Research

Two natural extensions of the model merit further investigation. First, the two-period framework could be extended to a multi-period setting with Bayesian learning about δ . In such a model, households update beliefs about the lake expansion trajectory over time, and the interaction between tenure security and belief updating would determine the optimal timing of adaptation versus migration. The slow-onset nature of Kenya’s rising lakes, with the creeping, acceleration, and plateau phases documented in Section 4, makes this setting

particularly amenable to a learning model, as households observe sequential signals about displacement severity.

Second, the partial-equilibrium framework abstracts from land market spillovers. As displaced households bid for land in buffer zones, prices rise, creating secondary displacement for land-insecure households in receiving areas. A general equilibrium extension could model these displacement cascades and their interaction with tenure formality – the prediction being that cascading effects are worse under informality, as land-insecure households in buffer zones are more easily displaced by incoming demand.

7 Conclusion

This paper develops and calibrates a model bridging two largely disconnected literatures: the economics of property rights and the economics of climate displacement. The key conceptual contribution is identifying the “land persistence” assumption embedded in the canonical property rights framework of Besley (1995) – the implicit assumption that land continues to exist. When environmental processes permanently destroy land, the standard channels (investment, credit, labor allocation) break down, and three displacement-specific channels become operative: compensation receipt, adaptation investment, and post-displacement reallocation. I formalize these channels and calibrate the model using satellite-derived lake expansion data from Kenya’s Rift Valley.

The calibrated model generates sharp quantitative predictions. Under an accounting decomposition, formally titled households recover 82 percent of baseline welfare versus 35 percent for informal occupants; under a consumption-equivalent metric, the figures are 92 percent versus 77 percent. Compensation accounts for 65 percent of the accounting gap. Pre-displacement titling programs yield benefit-cost ratios of 3:1 to 8:1. Tenure-blind compensation substantially narrows the formal-informal gap but cannot close it entirely, because adaptation and reallocation channels continue to operate through tenure security. Strong customary institutions provide partial insurance that can substitute for 40–70 percent of

the formal compensation gap. Displacement amplifies pre-existing gender inequality by a factor of $1.8\text{--}3.1\times$, with both titling and tenure-blind compensation operating as strongly gender-equalizing policies.

Several limitations warrant discussion. First, the model is deliberately partial-equilibrium: compensation $\bar{\kappa}$ is treated as exogenous, so the policy counterfactuals do not account for fiscal constraints on compensation budgets. If a universal titling program raised successful claims from 32 percent to 93 percent of displaced households, the aggregate compensation bill would increase substantially, potentially exceeding government capacity. The BCR estimates should be interpreted as upper bounds that assume the compensation supply adjusts. Second, the compensation channel dominates quantitatively (65 percent of the formal-informal gap), which raises the question of whether the three-channel framework is necessary. I argue that it is, for two reasons: the adaptation channel becomes more important under parameter values where compensation infrastructure is weak (the BCR falls from 5.4:1 to 1.1:1 without formal compensation), and the reallocation channel, while modest in the baseline, generates the model’s most novel prediction (that customary tenure can dominate formal tenure when compensation is equalized). A simpler one-channel model would miss both insights.

Third, tenure is treated as exogenous. If formal tenure yields substantially better recovery ($2.34\times$ in the accounting framework, $1.19\times$ in consumption-equivalent terms), one might ask why all households do not formalize. In the Kenyan context, formalization is constrained by: (i) historical allocation of community land under colonial and post-colonial tenure systems that predate individual titling; (ii) high fixed costs of surveying, registration, and legal fees relative to household income; and (iii) the value of community embeddedness that customary tenure provides, which the model captures through the reallocation and insurance channels. The policy counterfactual of titling programs can be understood as relaxing constraint (ii). Fourth, the two-period structure, while analytically tractable, does not fully exploit the slow-onset nature of the setting. The three displacement phases documented in Section 4 (creeping, acceleration, plateau) suggest that optimal household

behavior evolves over time as information about δ accumulates. The model captures the cumulative effect of slow-onset displacement but not the dynamic sequencing of decisions within it; the multi-period extension discussed in Section 6 would address this limitation.

Fifth, the functional form $p(s) = s^\alpha$ for the claims process is assumed rather than derived. Because compensation is the dominant channel, the results are sensitive to this specification. Table 3 shows that alternative forms (logistic, linear) preserve the qualitative ranking but affect magnitudes: the logistic specification, which concentrates the claims gradient around a threshold, produces a *larger* formal-informal gap than the baseline power form, while the linear form is intermediate. The power specification is therefore conservative for the formal-informal welfare gap. Nonetheless, the magnitudes of the BCR calculations should be treated as illustrative rather than precise pending empirical estimation of the claims function. Finally, the model validation relies partly on comparison with FEMA buy-out programs in the United States, which operate in a very different institutional context. While the recovery rate magnitudes are broadly consistent, this cross-context comparison should be interpreted cautiously; direct validation against Kenyan compensation data, when available, would strengthen the calibration.

Despite these limitations, the framework makes a clear contribution: it provides the first formal model connecting property rights to climate displacement outcomes, identifies the channels through which tenure shapes post-displacement welfare, and generates testable predictions that a companion empirical paper (Ipapa, 2026) can evaluate using household survey data from the Rift Valley. More broadly, the model suggests that the property rights agenda, long focused on investment and productivity in static settings, has important and previously unexplored implications for climate adaptation and disaster resilience.

A Mathematical Proofs for the Theoretical Framework

This appendix formalizes the theoretical framework presented in Section 2 and provides proofs for the key propositions. Section 3.1 states the optimization problem used for cali-

bration. I build from a standard household optimization problem, introduce environmental destruction, and derive the comparative statics underlying each hypothesis.

A.1 Setup and Definitions

Consider a household i endowed with a parcel of land of quality $q_i > 0$ and tenure status $T_i \in \{F, C, I\}$ (formal, customary, informal). Tenure status determines a security parameter $s_i = s(T_i) \in (0, 1]$, where $s(F) > s(C) > s(I)$. The parameter s_i captures the probability that the household retains recognized claims to its land and land-linked entitlements.

Time is discrete with two periods: $t = 0$ (pre-displacement) and $t = 1$ (post-displacement). At $t = 0$, the household chooses investment in productive capital k_i and adaptation capital a_i subject to a budget constraint. At $t = 1$, an environmental shock destroys a fraction $\delta_i \in [0, 1]$ of the land, and the household receives compensation and accesses reallocation opportunities.

Definition 1 (Tenure Security Function). *The tenure security function $s : \{F, C, I\} \rightarrow (0, 1]$ satisfies:*

1. $s(F) > s(C) > s(I)$
2. $s(T_i)$ is the probability that household i 's claims are recognized by formal institutions (government, courts, NGOs)

Definition 2 (Environmental Destruction). *An environmental shock at $t = 1$ destroys fraction $\delta_i \in [0, 1]$ of household i 's land. Remaining usable land is $(1 - \delta_i)q_i$. When $\delta_i = 1$, the land is fully destroyed (permanently submerged).*

A.2 Household Optimization Problem

At $t = 0$, the household solves:

$$\max_{k_i, a_i} U_i = u(c_i^0) + \beta \mathbb{E} [u(c_i^1)] \quad (42)$$

subject to the period-0 budget constraint:

$$c_i^0 = y_i - k_i - a_i - \phi(s_i) \quad (43)$$

where y_i is the household's period-0 endowment (normalized to 1 in the calibration) and $\phi(s_i) = \phi_0(1 - s_i)$ is a documentation cost decreasing in tenure security. The household discounts the future at rate $\beta \in (0, 1)$.

Period-1 consumption depends on displacement severity:

$$c_i^1 = f((1 - \delta_i \cdot \ell(a_i, s_i)) q_i, k_i) + g(a_i, s_i) + p(s_i) \bar{\kappa} \quad (44)$$

where $f(q, k) = Aq^{\alpha_q} k^{\alpha_k}$ is Cobb-Douglas production on effective land (reduced by displacement severity δ_i moderated by the asset protection function $\ell(a_i, s_i) = e^{-\lambda a_i s_i}$), and:

- $g(a_i, s_i) = \gamma a_i^\eta s_i$ captures returns to adaptation investment, with $g_a > 0$, $g_{aa} < 0$, and $g_{as} > 0$ (the key cross-partial)
- $p(s_i) \bar{\kappa}$ is expected compensation, with $p'(s) > 0$ (more secure tenure yields higher claims success)
- Reallocation $R_i(T_i, N_i)$ and community transfers $\Omega_i(\theta)$ enter the accounting welfare decomposition but are realized ex post and not part of the household's choice problem

A.3 Proof of Proposition 1: Compensation Channel

Proposition 1 (Compensation is Increasing in Tenure Security). *Suppose compensation κ_i is determined by a claims process in which the household must document ownership to receive payment. Let $p(s_i) \in [0, 1]$ denote the probability of a successful claim, with $p'(s) > 0$ and $p''(s) \leq 0$. Let $\bar{\kappa}(\delta_i, q_i)$ denote the full compensation value for documented losses. Then:*

1. *Expected compensation $\mathbb{E}[\kappa_i]$ is strictly increasing in s_i*
2. *The expected time to compensation receipt is strictly decreasing in s_i*

Proof. Part 1. Define expected compensation as:

$$\mathbb{E}[\kappa_i] = p(s_i) \cdot \bar{\kappa}(\delta_i, q_i) \quad (45)$$

Since $\bar{\kappa} > 0$ for any displaced household ($\delta_i > 0$) and $p'(s) > 0$ by assumption:

$$\frac{\partial \mathbb{E}[\kappa_i]}{\partial s_i} = p'(s_i) \cdot \bar{\kappa}(\delta_i, q_i) > 0 \quad (46)$$

Since $s(F) > s(C) > s(I)$, it follows that $\mathbb{E}[\kappa_i|T_i = F] > \mathbb{E}[\kappa_i|T_i = C] > \mathbb{E}[\kappa_i|T_i = I]$.

Part 2. Model compensation timing as a search process where the household submits documentation at cost $\phi(s_i)$, with $\phi'(s) < 0$ (formal documentation reduces bureaucratic cost). The household receives compensation at time τ_i , where τ_i is the first success in a sequence of claim attempts, each succeeding with probability $p(s_i)$. Thus τ_i follows a geometric distribution with parameter $p(s_i)$, giving:

$$\mathbb{E}[\tau_i] = \frac{1}{p(s_i)} \quad (47)$$

Since $p'(s) > 0$:

$$\frac{\partial \mathbb{E}[\tau_i]}{\partial s_i} = -\frac{p'(s_i)}{[p(s_i)]^2} < 0 \quad (48)$$

Expected wait time is strictly decreasing in tenure security. $\square$

A.4 Proof of Proposition 2: Adaptation Channel

Proposition 2 (Adaptation Investment is Increasing in Tenure Security). *Let the household choose adaptation investment $a_i \geq 0$ at $t = 0$. If the household expects to be displaced with probability $\pi_i \in (0, 1)$ and tenure security s_i determines the expected horizon of land use, then optimal adaptation a_i^* is strictly increasing in s_i .*

Proof. The household's expected utility from adaptation investment is:

$$V(a_i) = u(c_i^0(a_i)) + \beta \left[\pi_i u(c_i^{1,d}(a_i)) + (1 - \pi_i) u(c_i^{1,n}(a_i)) \right] \quad (49)$$

where $c_i^{1,d}$ is consumption if displaced and $c_i^{1,n}$ if not. Period-0 consumption is $c_i^0 = f(q_i, k_i) - k_i - a_i$, so $\partial c_i^0 / \partial a_i = -1$.

If displaced, adaptation reduces losses: $c_i^{1,d} = \tilde{f}_i - L(\delta_i, a_i) + \kappa_i + R_i$, where $L_a < 0$ and $L_{aa} > 0$ (diminishing returns to adaptation).

If not displaced, the household retains land with probability s_i and adaptation enhances productivity: $c_i^{1,n} = f(q_i, k_i) + g(a_i, s_i)$, where $g_a > 0$, $g_{aa} < 0$, and crucially $g_{as} > 0$ (the return to adaptation is higher when tenure is more secure, since the household expects to capture returns for longer).

The first-order condition for optimal a_i^* is:

$$u'(c_i^0) = \beta \left[\pi_i u'(c_i^{1,d}) \cdot (-L_a) + (1 - \pi_i) u'(c_i^{1,n}) \cdot g_a(a_i, s_i) \right] \quad (50)$$

The right-hand side is the marginal benefit of adaptation. Taking the cross-partial with respect to s_i :

$$\frac{\partial^2 V}{\partial a_i \partial s_i} = \beta (1 - \pi_i) u'(c_i^{1,n}) \cdot g_{as}(a_i, s_i) > 0 \quad (51)$$

since $g_{as} > 0$, $u' > 0$, and $\beta, (1 - \pi_i) > 0$.

By the implicit function theorem applied to the first-order condition, and noting that the second-order condition $V_{aa} < 0$ holds (by concavity of u and $L_{aa} > 0$, $g_{aa} < 0$):

$$\frac{da_i^*}{ds_i} = -\frac{V_{as}}{V_{aa}} > 0 \quad (52)$$

since $V_{as} > 0$ and $V_{aa} < 0$. Optimal adaptation investment is strictly increasing in tenure security. $\square$

Proposition 3 (Adaptation Reduces Displacement Losses). *Asset losses from displacement*

are strictly decreasing in adaptation investment: $\partial L / \partial a_i < 0$.

Proof. Model asset losses as $L(\delta_i, a_i) = \delta_i \cdot h(q_i, k_i) \cdot \ell(a_i)$, where $h(q_i, k_i)$ is the value of assets at risk and $\ell(a_i) \in (0, 1]$ is the fraction of at-risk assets destroyed, with $\ell'(a) < 0$ (adaptation protects assets) and $\ell''(a) > 0$ (diminishing protection). Then:

$$\frac{\partial L}{\partial a_i} = \delta_i \cdot h(q_i, k_i) \cdot \ell'(a_i) < 0 \quad (53)$$

Combining with Proposition 2:

$$\frac{dL}{ds_i} = \frac{\partial L}{\partial a_i} \cdot \frac{da_i^*}{ds_i} < 0 \quad (54)$$

Households with more secure tenure invest more in adaptation, which reduces displacement losses. The total effect of tenure security on losses is unambiguously negative. $\square$

A.5 Proof of Proposition 4: Reallocation Channel

Proposition 4 (Ambiguous Effect of Tenure Formality on Reallocation). *Let reallocation value R_i have two components: formal reallocation $R_i^F(s_i)$ and community-based reallocation $R_i^C(N_i, T_i)$. Then the effect of moving from customary to formal tenure on total reallocation is ambiguous.*

Proof. Define total reallocation value as:

$$R_i = R^F(s_i) + R^C(N_i, T_i) \quad (55)$$

where R^F captures value from government relocation programs, legal claims to new land, and sale of residual rights, while R^C captures value from community-based land sharing, elder-mediated reallocation, and kinship networks.

For formal reallocation: $\partial R^F / \partial s_i > 0$, since formal programs require documentation and recognized claims.

For community reallocation, the key insight is that customary tenure embeds the household in reciprocal obligations that formal title may dissolve. Specifically, assume $R^C(N_i, C) > R^C(N_i, F)$ when community institutions are strong (high N_i), because:

1. Formally titled households may have exited communal land-sharing arrangements
2. Community reallocation prioritizes members with recognized customary claims
3. Formalization may signal exit from the communal risk-sharing network

The total effect of moving from $T_i = C$ to $T_i = F$ is:

$$\Delta R_i = \underbrace{[R^F(s(F)) - R^F(s(C))]}_{>0} + \underbrace{[R^C(N_i, F) - R^C(N_i, C)]}_{<0 \text{ when } N_i \text{ is high}} \quad (56)$$

The sign of ΔR_i depends on the relative magnitudes, which vary with institutional context. When formal institutions are strong and community networks weak, $\Delta R_i > 0$. When community institutions are strong and formal mechanisms absent or slow, $\Delta R_i < 0$. □

A.6 Proof of Proposition 5: Gender Heterogeneity

Proposition 5 (Compounding Gender Disadvantage). *Let $\sigma_i \in \{M, W\}$ index gender of household head. If women face systematically lower tenure security $s(T_i|W) < s(T_i|M)$ for all T_i , and if displacement can sever tenure claims held through marriage, then gender gaps in post-displacement welfare exceed gender gaps in pre-displacement welfare.*

Proof. Define the gender gap in post-displacement welfare as:

$$\Delta W^{post} \equiv \mathbb{E}[W^{post}|M] - \mathbb{E}[W^{post}|W] \quad (57)$$

From equation (1), post-displacement welfare depends on three channels. I show the gender gap is positive and larger than pre-displacement gaps in each.

Channel 1 (Compensation): By Proposition 1, $\mathbb{E}[\kappa_i]$ is increasing in s_i . Since $s(T|W) < s(T|M)$:

$$\mathbb{E}[\kappa|M] - \mathbb{E}[\kappa|W] = p'(\tilde{s}) \cdot \bar{\kappa} \cdot [s(T|M) - s(T|W)] > 0 \quad (58)$$

for some $\tilde{s}$ by the mean value theorem.

Channel 2 (Adaptation): By Proposition 2, a_i^* is increasing in s_i , so $a^*(M) > a^*(W)$. By Proposition 3, this implies $L(M) < L(W)$: women-headed households suffer larger displacement losses.

Channel 3 (Reallocation): In patrilineal systems, let $\mu \in (0, 1)$ be the probability that displacement severs a woman's tenure claim held through marriage (due to spouse death, household dissolution, or return to natal community). Then women's effective reallocation value is:

$$\mathbb{E}[R|W] = (1 - \mu)R(T_i, N_i) + \mu \cdot R(I, N'_i) \quad (59)$$

where $N'_i \leq N_i$ reflects potentially weaker networks after household dissolution, and the second term uses $T = I$ because severed marital claims revert to informal status. Since $R(I, N') < R(T, N)$:

$$\mathbb{E}[R|M] - \mathbb{E}[R|W] = \mu [R(T, N) - R(I, N')] > 0 \quad (60)$$

Combining all three channels:

$$\Delta W^{post} = \underbrace{\Delta \kappa}_{>0} + \underbrace{(-\Delta L)}_{>0} + \underbrace{\Delta R}_{>0} + \underbrace{\Delta W^{pre}}_{\text{baseline gap}} \quad (61)$$

The pre-displacement gender gap ΔW^{pre} operates only through the standard property rights channels (investment, credit, labor). Post-displacement, three additional gap-widening terms emerge. Therefore $\Delta W^{post} > \Delta W^{pre}$: displacement amplifies pre-existing gender inequality. $\square$

A.7 Proof of Proposition 6: Collective Tenure and Partial Insurance

Proposition 6 (Community Insurance Under Collective Tenure). *Suppose households with customary tenure belong to a community that operates a mutual insurance mechanism. Let the community pool resources and redistribute to displaced members. Then customary tenure provides partial insurance against displacement losses, with the degree of insurance increasing in institutional quality.*

Proof. Let $\theta \in [0, 1]$ index the quality of collective institutions (governance, enforcement, norms). Consider a community of n households, of which $n_d \leq n$ are displaced. The community operates a transfer scheme where non-displaced members contribute τ to displaced members.

Each non-displaced member's contribution satisfies participation constraint:

$$u(c^{1,n} - \tau) + \theta \cdot V^{cont} \geq u(c^{1,n}) \quad (62)$$

where V^{cont} is the continuation value of remaining in the insurance arrangement (capturing future reciprocity). This yields maximum sustainable transfer:

$$\tau^*(\theta) = c^{1,n} - u^{-1}(u(c^{1,n}) - \theta V^{cont}) \quad (63)$$

Since u is concave, τ^* is increasing in θ : $\partial \tau^* / \partial \theta > 0$.

Total community transfer to each displaced household is:

$$\Omega_i = \frac{(n - n_d)}{n_d} \cdot \tau^*(\theta) \quad (64)$$

For a household with customary tenure, effective compensation becomes:

$$\kappa_i^{eff} = \kappa_i(s(C)) + \Omega_i(\theta) \quad (65)$$

Compare with a formally titled household outside the community network:

$$\kappa_j^{eff} = \kappa_j(s(F)) \quad (66)$$

The customary household's effective compensation exceeds the formal household's when:

$$\Omega_i(\theta) > \kappa_j(s(F)) - \kappa_i(s(C)) = p'(\bar{s})\bar{\kappa}[s(F) - s(C)] \quad (67)$$

This holds when institutional quality θ is sufficiently high and the formal-customary security gap $s(F) - s(C)$ is not too large. Formally, there exists a threshold $\bar{\theta}$ such that for $\theta > \bar{\theta}$, community insurance fully compensates for the formal compensation gap, providing partial (and potentially full) substitution. $\square$

A.8 Welfare Decomposition

Combining the results above, I derive a complete decomposition of post-displacement welfare.

Proposition 7 (Welfare Decomposition). *Post-displacement welfare admits the decomposition:*

$$W_i^{post} = \underbrace{W_i^{pre}}_{\text{baseline}} - \underbrace{L(\delta_i, a_i^*(s_i))}_{\text{losses (decreasing in } s_i)} + \underbrace{\kappa_i(s_i)}_{\text{compensation (increasing in } s_i)} + \underbrace{R_i(T_i, N_i)}_{\text{reallocation (ambiguous in } T_i)} \quad (68)$$

where $a_i^*(s_i)$ is optimal adaptation from Proposition 2. The total effect of tenure security on post-displacement welfare satisfies:

$$\frac{dW_i^{post}}{ds_i} = -L_a \cdot \underbrace{\frac{da^*}{ds_i}}_{>0} + \underbrace{\kappa_s}_{>0} + \underbrace{\frac{\partial R}{\partial s_i}}_{\geq 0} > 0 \quad (69)$$

when the compensation and loss-reduction channels dominate the potentially negative reallocation effect. This condition holds when formal compensation mechanisms exist (κ_s

is bounded away from zero) and community institutions do not fully substitute for formal claims.

Proof. The first term follows from Propositions 2 and 3: higher s_i increases a_i^* , which decreases L_i , so the contribution is positive. The second term is positive by Proposition 1. The third term is ambiguous by Proposition 4. The sum is positive when:

$$-L_a \cdot \frac{da^*}{ds_i} + \kappa_s > \left| \frac{\partial R}{\partial s_i} \right| \cdot \mathbf{1} \left[\frac{\partial R}{\partial s_i} < 0 \right] \quad (70)$$

This holds generically because the reallocation penalty from formalization is bounded (community transfers Ω are bounded by community resources), while compensation and loss-reduction benefits are proportional to household asset values and displacement severity. $\square$

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